

FROM EVIDENCE TO JUDGMENT: TRACE-AUGMENTED REASONING UNDER CONFLICTING EVIDENCE

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ABSTRACT

Retrieval-augmented generation (RAG) lets a language model answer a question by first retrieving relevant documents and then writing an answer grounded in them. In practice, however, retrieved documents often disagree with each other, contain outdated information, or fail to support any answer at all. Current systems have no reliable way to judge which of these documents to trust before responding. We argue that this is a problem of judgment, not of answer generation, and we test the two abilities separately. To do this, we release a benchmark of 1,679 examples covering 10,343 retrieved, labeled documents, and we fine-tune open-weight (publicly released) language models to produce a structured evidence trace before answering. This trace checks each source individually, analyzes how the sources conflict, and produces a final answer or a justified refusal. We evaluate the results with a new framework we call CATS (Conflict-Aware Trust Score), which scores four aspects of a response: whether the model correctly declines to answer when no source supports one (grounded refusal), how consistently it follows that decision across prompt styles (behavioural adherence), whether the cited documents back its claims (factual grounding), and whether it still finds the right answer when one exists (single-truth recall). We validate the latter three against an independent human study and compute grounded refusal deterministically. Trace supervision substantially improves the refusal decision. Refusal F1 rises from as low as 0.429 without it to at least 0.826 with it, across every model and prompt setting we test. Trace-tuned models are also far more consistent across different prompt phrasings, while baseline behavior swings widely. Answer quality itself does not improve: single-truth recall is unchanged, and the grounding advantage we see under well-specified prompts disappears once we correct a measurement artefact we identify. Overall, structured evidence traces make conflict-aware refusal reliable and prompt-robust without improving the model’s ability to find correct answers.

1 INTRODUCTION

Retrieval-augmented generation conditions a generator on a small set of retrieved snippets, ideally with faithful attribution (Lewis et al., 2020; Borgeaud et al., 2022; Gao et al., 2024; Guu et al., 2020). In practice, retrieval rarely returns a coherent evidence set, instead producing documents that vary in relevance, quality, credibility, and recency. The retrieved documents may contain conflicting claims, provide only partial support, or collectively be insufficient to answer the user’s query.

Such a set demands a sequence of judgments before the model writes a single word of its answer. These judgments are which document says what, how the documents relate to one another, and whether what remains is sufficient. A system that treats the snippets as interchangeable context produces a fluent, confidently wrong answer. A system that refuses whenever sources disagree is no better, since disagreement is often compatible with a properly scoped answer.

Existing work addresses parts of this, but each stops short of a different part. *DRAGged into Conflicts* (Cattan et al., 2025) supervises the conflict label rather than the reasoning that produces it. *Chain-of-Note* (Yu et al., 2024) produces per-document notes without characterising how the retrieved set coheres. *Trust-Score/Trust-Align* (Song et al., 2025) score refusal and citation behaviour without

Code, benchmark, and results are available at an anonymous repository: https://anonymous.4open.science/r/rag_reason-2F8F/.

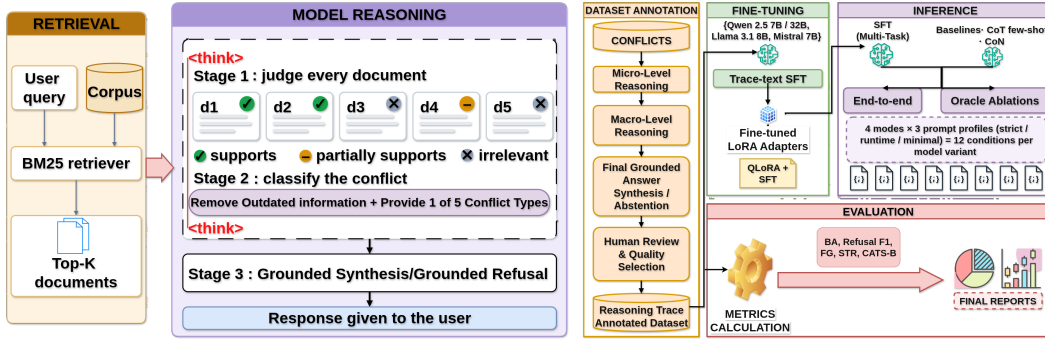


Figure 1: The evidence trace. The model adjudicates each retrieved document, infers the cross-document conflict pattern, and produces a conflict-conditioned grounded answer or a justified refusal; **Figure 2:** System overview. A weighted multi-annotator committee produces the supervision corpus. Models are fine-tuned to emit the three-stage trace. CATS evaluates grounded refusal, behavioural adherence, factual grounding, and single-truth recall, with an independent human study validating the latter three. We compute grounded refusal deterministically.

conditioning the expected response on the kind of conflict present. Closest in framing, Wu et al. (2026) also fine-tune open models on conflict signals, but derive those signals from generation probabilities rather than from an explicit textual record. Wang et al. (2025) resolve conflicting evidence by inference-time multi-agent debate rather than supervision. Wang et al. (2026) pair a learned conflict detector with principled credibility scoring, where our source-quality field is a coarse URL heuristic. Muhamed et al. (2025) evaluate selective refusal under flawed context and independently find it to be a trainable capability, which is the deficit our results localise. The gap that remains is a single framework that both supervises and evaluates per-document reasoning, cross-document conflict inference, and justified abstention together.

We close that gap with a *structured evidence trace*. Before answering, the model exposes a three-stage record of its own adjudication (Figure 1). **Stage 1** addresses which document says what: it assigns each snippet a verdict with an extracted key fact and a source-quality label. **Stage 2** addresses how the documents relate to one another: it aggregates those judgments into one of five conflict patterns, following the taxonomy of Cattani et al. (2025). **Stage 3** addresses whether what remains is sufficient to answer: it produces a citation-linked answer or a justified abstention, conditioned on that pattern (§3). The model emits the trace under a fixed output contract rather than as free-form reasoning. This makes it machine-checkable, so a failure is attributable to a stage rather than to the system as a whole. Liu et al. (2026) share this philosophy, training auditable traces by reinforcement learning on multi-hop QA rather than supervising conflict adjudication. Prompting can produce conflict-aware answers, but it cannot produce conflict-aware judgment. Separating the two empirically is this paper’s central test.

A benchmark that can carry the claim. Testing this requires evaluation data that is genuinely held out. We built an independent 736-example benchmark with *zero* identifier or normalised-query overlap with training, verified on the released files rather than merely asserted (a recipe-development caveat on this disjointness is in Appendix D). Because open-web retrieval cannot surface some conflict regimes, a curated track supplies them, and we report its composition per class (§4).

Trace supervision’s clearest gain is the refusal decision: F1 rises from as low as 0.429 without it to at least 0.826 with it, and this gain holds regardless of how we phrase the prompt, unlike baseline models, whose behavior swings widely. The improvement lies in judgment, not answer generation, since single-truth recall itself does not improve.

Contributions. (1) **A conflict-aware RAG resource (§4):** 943 training/validation examples and a disjoint 736-example benchmark. Each of the 10,343 adjudicated evidence units carries a verdict, key fact, snippet-anchored quote, rationale, source-quality label, and the full annotator vote tally. (2) **Evidence that trace supervision buys judgment, not answer quality (§8):** refusal calibration and prompt-invariance improve, but single-truth recall and grounding under well-specified prompts do not. (3) **CATS (§6):** an evaluation framework that scores a conflict-aware response as four separate components rather than one number, each reported over its own applicability denominator. We

validate three of the four components against an independent human study and compute the fourth, grounded refusal, deterministically. **(4) Oracle ablations (§8.4):** supplying gold information one stage at a time localises which stage limits the system and gives construct-validity evidence for the three-stage decomposition.

2 RELATED WORK

2.1 KNOWLEDGE AND EVIDENCE CONFLICTS

We scope this paper to conflicts *among retrieved documents*. A model’s parametric knowledge is the information it memorized during training, as opposed to what it retrieves at inference time. Conflict between that knowledge and retrieved evidence, the context–memory setting targeted by much recent work (Huo et al., 2025; Zhang et al., 2025; Ye et al., 2026), is out of scope here. We evaluate our systems only on whether they reason correctly over the supplied set. Retrieval-augmented generation conditions a model on corpus-retrieved documents at generation time (Guu et al., 2020; Lewis et al., 2020; Izacard et al., 2023; Borgeaud et al., 2022; Gao et al., 2024; Ram et al., 2023). Retrieved documents routinely carry conflicting information, which complicates generation and has become a research area of its own (Xu et al., 2024a). *ConflictBank* (Su et al., 2024) evaluates how models handle conflicts within retrieved knowledge, within parametric knowledge, and in their interplay. It pairs Wikidata claims with LLM-generated contradictory evidence across multiple textual styles. *WikiContradict* (Hou et al., 2024) benchmarks robustness to explicit and implicit contradictions between Wikipedia passages, with human-annotated instances. *ECON* (Jiayang et al., 2024) studies detection and resolution of evidence conflicts directly. *DRAGged into Conflicts* (Cattan et al., 2025) is closest to our setting. It proposes a conflict taxonomy spanning temporal, debatable, misinformation, and complementary cases. It demonstrates that models often fail to adapt their responses without explicit conflict-aware guidance. Most of this literature focuses on *detecting* or *classifying* conflict. Our framework instead supervises the reasoning that produces the classification. It evaluates whether a model’s document-level judgments, conflict inference, and final response behaviour are mutually consistent, rather than scoring only the final answer.

2.2 REASONING SUPERVISION, MULTI-HOP AND ROBUSTNESS IN RAG

Several lines of work supervise or elicit intermediate reasoning. *Chain-of-Thought* prompting established that generating intermediate steps improves complex reasoning (Wei et al., 2022). *Chain-of-Note* (Yu et al., 2024) adapts this to retrieval by generating sequential reading notes for each retrieved document. This improves robustness to noisy and irrelevant passages and lets a model answer “unknown” when neither retrieved nor intrinsic knowledge suffices. Adaptive adversarial training (Fang et al., 2024) and robustness training (Yoran et al., 2024) mitigate retrieval noise, and *GSM-IC* (Shi et al., 2023) shows that irrelevant context substantially degrades reasoning. Our work differs in two respects. First, each stage has a closed label set, so Stage-1 and Stage-2 decisions are separately scorable against gold annotations rather than merely inspectable. Second and more importantly, *we treat the reasoning trace as an object of evaluation rather than a means of improving the answer*. The three-stage decomposition earns its place only if the stages are genuinely separable. We test this directly by supplying gold information for one stage at a time. We then ask whether each intervention moves only the metrics that stage should govern (§8.4).

2.3 MEASURING TRUST, GROUNDEDNESS AND CONFLICT ALIGNMENT

Trustworthiness metrics for RAG largely score an answer against the retrieved set as a whole. Conflict-aware evaluation instead requires that the *expected* response depend on the kind of disagreement present. A single scoring rule should not treat a balanced summary and a decisive correction the same way. Evaluations of RAG frequently assess end-to-end system quality (Gao et al., 2024; Xu et al., 2024b), conflating retriever quality with generator capability (Fan et al., 2024). *Trust-Score* (Song et al., 2025) advances this by evaluating trustworthiness through answer correctness, citation reliability, and refusal behaviour. Its companion *Trust-Align* aligns models to improve that score. *Trust-Align*’s held-out refusal material forms the refusal track of our resource. *FaithEval* (Ming et al., 2025) probes whether models stay faithful to context under adversarial or counterfactual conditions. CATS extends this line in three ways. It conditions the expected response on the inferred conflict type. It evaluates grounding at the claim level with citation sensitivity. It also reports every component over

its own applicability denominator, since the four constructs apply to different subsets of examples. Conflating them would confound the comparison the framework aims to support (§8.3). We validate the judges against an independent human study (Zheng et al., 2023) and report which components that validation supports and which it does not (§9).

3 TASK FORMULATION AND THE EVIDENCE TRACE

An example consists of a query q and a retrieved set $D = \{d_1, \dots, d_N\}$. The system must produce either a citation-linked answer or the controlled refusal string `CANNOT ANSWER`, `INSUFFICIENT EVIDENCE`. It reasons only over the retrieved evidence and refuses when that evidence is insufficient. Rather than mapping (q, D) directly to a response, we supervise an intermediate representation emitted as part of the output.

Stage 1: document adjudication. Every d_i receives a verdict in $\{\textit{supports}, \textit{partially supports}, \textit{irrelevant}\}$, a one-sentence key fact entailed by the snippet, and a coarse source-quality label. The *partially supports* class is deliberately broad, covering on-topic but incomplete, hedged, scoped, or single-facet evidence. This choice avoids the failure mode we care about: a binary useful/useless split causes refusal whenever no single snippet is individually decisive.

Stage 2: conflict analysis. The Stage-1 judgments are aggregated into exactly one of five labels, each prescribing a different aggregation strategy. Table 19 (Appendix G) gives each type’s definition and the response behaviour it requires. The taxonomy separates *complementarity* from *conflict*: two snippets can contribute different necessary pieces without disagreeing.

Stage 3: grounded synthesis. Stages 1 and 2 are emitted *inside* the `<think>` block. Stage 3 is the response itself and sits outside it. The final response is conditioned on the conflict label, with sentence-level [dX] citations, or is the exact refusal string, terminating with [[END-OF-ANSWER]]. Figure 2 places this in the overall pipeline.

4 DATASET GENERATION AND VALIDATION

Table 1 summarizes the generated dataset and its validation. Appendix A gives the complete construction record, per-source distributions, and retrieval protocol; Appendix C gives the human-review statistics.

4.1 TRAINING AND VALIDATION

The 943-example training release (862 train / 81 validation) combines three source families: **CONFLICTS** (Cattan et al., 2025) and held-out refusal examples derived from Trust-Align (Song et al., 2025). The third family draws on examples from the benchmark construction stream that were *not* admitted to the final benchmark. We checked these examples for identifier overlap against the released benchmark and found the intersection empty. Appendix A gives the per-family counts.

4.2 AN INDEPENDENT BENCHMARK

The 736-example benchmark is built in three tracks and reporting them separately is deliberate. **(1) Track A: web-retrieved (508).** Queries are drawn from ConflictingQA (Wan et al., 2024), SituatedQA (Zhang & Choi, 2021), FreshQA (Vu et al., 2024) and QACC (Liu et al., 2025), with every query overlapping the training pool excluded. Documents are retrieved live and a query-focused window is selected by a TAS-B dense bi-encoder (Hofstätter et al., 2021). Appendix A gives the full funnel and the five-document sampling rule. **(2) Track B: refusal (128).** Refusal-required examples come from the held-out refusal pool of Trust-Align (Song et al., 2025) and are *not* web-retrieved. **(3) Track C: curated conflict (100).** Open-web retrieval cannot reliably surface every conflict regime, such as a fact whose truth value has since changed. We therefore add 78 examples built from dated Wikipedia revision pairs, in the spirit of Hou et al. (2024), 19 multi-hop contexts from HotpotQA (Yang et al., 2018) and three medical/misinformation cases. The track is strongly class-targeted, cleanly separating outdated-information and complementary cases (Table 5, Appendix A).

Conflict type	Train		Benchmark		
	<i>n</i>	%	<i>n</i>	%	
No conflict	343	36.4	211	28.7	Grounded Refusal (GR) whether the system answered exactly when the evidence permits answering; scored deterministically from the answerability label
Complementary	286	30.3	221	30.0	
Conflicting opinions	158	16.8	109	14.8	Behaviour Adherence (BA) whether the answer followed the response policy its conflict type requires
Outdated	136	14.4	158	21.5	
Misinformation	20	2.1	37	5.0	Factual Grounding (FG) whether each extracted claim is both supported by an eligible document <i>and</i> cited
Answerable	649	68.8	608	82.6	
Refusal-required	294	31.2	128	17.4	Single-Truth Recall (STR) whether the response adopts the target answer as its own conclusion, rather than quoting or attributing it
Records	943		736		
Retrieved documents	6,642		3,701		
Docs / record (mean)	7.04		5.03		
Unique source domains	2,633		1,431		

Table 1: Dataset composition and scale. Conflict-type and answerability distribution of the training split (943 records, including validation) and the held-out benchmark (736). The benchmark is deliberately harder than the training distribution. It has fewer no-conflict cases (28.7% against 36.4%), a $1.5\times$ larger outdated-information share, and a $2.4\times$ larger misinformation share. Full per-source composition in Appendix A; **Table 2: The four CATS components**, each reported over its own applicability denominator because the constructs apply to different subsets of examples. Appendix G gives the full definitions, FG eligibility rules, per-type BA policies and the committee voting rule.

4.3 ANNOTATION DEPTH & HUMAN REVIEW

Each of the 10,343 retrieved documents carries six fields: an adjudicated *verdict*, an extracted *key fact*, and a snippet-anchored *supporting quote*. It also carries a *rationale*, a *source-quality label*, and the complete *annotator vote tally*. Labels come from a weighted multi-annotator committee that votes on the categorical decision at each stage. The committee then adopts the winning side’s explanation bundle intact from its highest-weight member, keeping each verdict, quote, and rationale internally coherent. The annotation contract requires the key fact to be entailed by its quote. It also requires the quote to be a contiguous verbatim span of at most fifty words. Appendix B gives the formal aggregation rule and the full contract. The prompts that enforce it are released with the resource (Appendix N).

Two human workflows, kept separate, cover the two populations. Seven reviewers screened the benchmark candidates on a retention decision, a preliminary conflict type, and eight quality judgments (Appendix C lists each). This screening reached 94.77% five-way conflict-type agreement (Cohen’s $\kappa = 0.92$) and 98.21% retention agreement ($\kappa = 0.92$). Because this second pass is *sequential*, the figures measure verification consistency rather than blind double annotation. Training-side review (two reviewers per example) reached 83.46% conflict-type agreement ($\kappa = 0.77$). A separate, class-balanced 100-pair study checks the Stage-1 per-document verdict itself, not the downstream conflict-type label. Three reviewers agree at Fleiss’ $\kappa = 0.83$ and agree with the committee at 92% ($\kappa = 0.88$ against the human majority). Appendix C and Appendix C.1 give the full protocols.

5 TRACE-SUPERVISED FINE-TUNING

5.1 OBJECTIVE AND PROMPT VARIANTS

The supervision target is the serialised trace followed by the final response. Three prompt variants are used: the **strict** prompt states the full taxonomy, verdict heuristics, and output contract (1,965 words). The **runtime** prompt keeps a compact structural schema (195 words). The **minimal** prompt supplies only the query, the documents, and a four-sentence evidence policy (40 words). The mixture also supervises document-verdict, conflict-type, and final-response prediction as separate task views. This gives each component explicit signal in isolation as well as inside the full trace. We train and evaluate across all three variants to test whether conflict-aware behaviour is *learned* or merely *recited* from the instruction (§8.2). Appendix O gives the three prompts verbatim and their design rationale.

5.2 FINE-TUNING

All models are fine-tuned with QLoRA (Dettmers et al., 2023; Hu et al., 2021), a parameter-efficient method that trains low-rank adapters (LoRA) atop a frozen, quantized base model. Training uses the standard token-level cross-entropy of the trace and response, with no auxiliary loss. Appendix D gives the hyperparameters, the checkpoint selection criterion, and the two training mixtures (recipes K and L). These mixtures differ by model family and are needed to interpret the per-model results in §8.

6 CATS: CONFLICT-AWARE TRUST SCORE

Standard RAG metrics collapse distinct failures. Answering when evidence is absent, mishandling the conflict type, making unsupported claims, and omitting the target answer are all “a bad answer”. CATS separates them into four components with independent applicability.

Table 2 states what each component measures and the population it is scored over. The definitions are operational. FG counts a claim only when the supporting document’s *gold* verdict is *supports* or *partially supports*. STR excludes conflicting-opinion items because no single conclusion exists for them. Appendix G sets out each rule. The denominators differ, so any comparison against another framework should start there.

Judges: BA, FG, and STR are judged by a committee of three locally served open-weight models with fixed weights, aggregated by weighted majority. Factual-grounding support additionally requires corroboration by at least two judges, and GR is never committee-judged. We compare this against human review in §9. The complete judge bundle is one system message plus three task prompts and the five-type behaviour rubric, reproduced verbatim in Appendix P. Metric definitions and aggregation are in Appendix G. **Aggregate scores:** the four components can also be combined into a single number per example, gated by the answer/refuse decision so that a wrong decision scores zero. We define two averages, **CATS-Prevalence** and **CATS-Balanced**, in Appendix G.8. We do not report either in the main results. Because a refusal-required example is scored on its decision alone, such examples carry 40% of CATS-Balanced while making up 17.4% of the benchmark. A model can therefore raise the aggregate by improving refusals without improving any answer. Every claim in §8 is stated on the components.

7 EXPERIMENTAL SETUP

7.1 PRIMARY EVALUATION

We fine-tune four instruction-tuned open-weight models, Qwen2.5-7B and Qwen2.5-32B (Yang et al., 2024), Llama-3.1-8B (Llama Team, AI at Meta, 2024) and Mistral-7B (Jiang et al., 2023), and compare each against its baseline version on the four CATS components. Within a cell, the benchmark examples, prompts, decoding policy, retry policy, post-processing, and evaluator are identical. Only the LoRA adapter differs (Table 3). Each pair is evaluated under the strict and minimal prompts, with the intermediate runtime prompt reported in Appendix R. Generation is deterministic (temperature 0) with profile-specific continuation budgets and sentinel-based stopping. Retries only enlarge the token budget for the identical prompt, never resampling or repairing content (Appendix F). Per-example scores lie in $[0, 1]$, so a distribution-free 95% interval half-width at $n = 736$ is ± 0.034 . We treat differences below this threshold as uninformative. We report exact sign tests for paired comparisons across the twelve model $\times$ prompt cells, and use exact Clopper–Pearson (binomial) intervals on the 128-example refusal subgroup.

7.2 ABLATION STUDIES

Oracle ablations. To test whether the three stages are separable, we re-run the fine-tuned models with gold information injected one stage at a time (§8.4). The **+conflict label** condition supplies the gold Stage-2 type, and **+per-doc notes** supplies the gold Stage-1 verdicts and key facts. The **+both** condition supplies each. **Comparison with prompting methods.** *Chain-of-Note* (Yu et al., 2024) writes per-document reading notes before answering, the closest analogue to our Stage 1. *Few-shot Chain-of-Thought* (Wei et al., 2022) supplies worked exemplars of the reasoning format. Both receive the identical benchmark, decoding policy, and evaluator (§8.5).

8 RESULTS

Table 3 reports the end-to-end matrix, each metric as a base/SFT column pair. Appendix R gives all twelve model×prompt cells.

Model	Prompt	Refusal F1		Behaviour Adherence		Factual Grounding		Single-Truth Recall	
		base	SFT	base	SFT	base	SFT	base	SFT
Qwen2.5-7B	strict	0.429	0.911	0.811	0.780	0.617	0.838	0.561	0.687
	minimal	0.031	0.881	0.737	0.768	0.031	0.837	0.757	0.686
Qwen2.5-32B	strict	0.367	0.905	0.789	0.821	0.851	0.812	0.733	0.749
	minimal	0.015	0.911	0.747	0.809	0.108	0.822	0.773	0.738
Llama-3.1-8B	strict	0.255	0.926	0.721	0.805	0.705	0.852	0.634	0.727
	minimal	0.015	0.934	0.743	0.795	0.074	0.857	0.710	0.714
Mistral-7B	strict	0.352	0.826	0.660	0.793	0.624	0.796	0.563	0.644
	minimal	0.000	0.839	0.727	0.786	0.079	0.792	0.700	0.627

Table 3: **End-to-end results on the 736-example held-out benchmark.** Within each pair the benchmark, prompts, decoding policy and evaluator are identical. Only the LoRA adapter differs. Bold marks a fine-tuned value that exceeds its baseline. Differences below the ± 0.034 noise floor (§7) are uninformative. We show the two endpoints of the instruction ladder. The intermediate runtime condition is in Appendix R.

Model	Method	Ref. F1	Behaviour Adherence	Factual Grounding	Single-Truth Recall
Llama-3.1-8B	CoN (prompt-only)	0.000	0.632	0.609	0.658
	CoT (few-shot)	0.302	0.716	0.635	0.678
	TRACE-SFT (ours)	0.926	0.805	0.852	0.727
Mistral-7B	CoN (prompt-only)	0.000	0.685	0.522	0.676
	CoT (few-shot)	0.525	0.657	0.417	0.528
	TRACE-SFT (ours)	0.826	0.793	0.796	0.644
Qwen2.5-7B	CoN (prompt-only)	0.000	0.727	0.666	0.719
	CoT (few-shot)	0.402	0.781	0.734	0.724
	TRACE-SFT (ours)	0.911	0.780	0.838	0.687

Table 4: **External prompting comparators** on baseline models; trace-tuned rows use the strict prompt, so every row receives a well-specified instruction. Both comparators reach usable answer content but neither acquires the answer/refuse decision.

8.1 REFUSAL RECALL IMPROVES SUBSTANTIALLY

The largest and cleanest effect is the answer/refuse decision. Refusal F1 separates completely: every baseline configuration scores ≤ 0.429 and every trace-tuned configuration ≥ 0.826 , a mean improvement of $+0.709$. This is not merely over-caution: decision accuracy also rises in every cell. The benchmark is 82.6% answerable, so answering everything scores 0.826 for free. No baseline configuration exceeds that floor by more than the ± 0.034 noise interval, while every trace-tuned cell reaches 0.927–0.976. Behaviour Adherence also improves, but unevenly, in an order that tracks baseline instruction-following (Table 3; recipes differ by model family, Appendix S). Appendix R gives the per-cell recall and accuracy figures, including the Llama-3.1-8B case, where decision accuracy masks near-zero refusal recall.

8.2 LEARNT BEHAVIOUR IS PROMPT-INVARIANT

Removing the prompt scaffolding is close to a null operation for fine-tuned models and close to catastrophic for baseline ones. The minimal condition is a sparse instruction, forty words that never

name the trace, the taxonomy or the citation format. A model that still produces them is therefore reproducing a learned mapping rather than following the prompt. Component by component, no trace-tuned model varies by more than the ± 0.034 noise floor across the ladder. Baseline spreads are 0.240–0.398 on refusal F1 and 0.620–0.823 on factual grounding. Under the minimal prompt, baseline grounding collapses toward zero while trace-tuned models hold steady (Table 26). The result also reproduces on citation measurements built after and independently of CATS. Figure 13 plots the full ladder. Appendix K gives the remaining figures.

8.3 WHAT TRACE SUPERVISION DOES *not* BUY

Two components show no consistent gain. Single-truth recall is positive in only 7 of 12 cells (sign test $p = 0.77$): fine-tuned models are not recovering target answers more often. Factual grounding is not directly comparable across run types, because it is averaged over $fg_n = 608 + \text{missed refusals}$. A baseline model that rarely refuses is therefore additionally scored on ~ 130 items whose evidence is insufficient by construction. Bounding FG on the like-for-like population, the intervals overlap in 6 of 12 cells, every one of them a strict or runtime cell (Table 25; see Appendix J). Trace supervision therefore raises grounding when the instruction is sparse, but under a well-specified prompt no consistent advantage is established. Given a well-specified prompt, a baseline model already produces competitive grounded answers. What it lacks is the judgment to decide whether to answer at all, and the ability to retain any of this without being told the procedure. That is what trace supervision supplies, which is why these negative results are predictions of our account rather than exceptions to it. Appendix J gives the per-cell bounds and the instruction-volume effect for baseline models.

8.4 MECHANISM-MATCHED ORACLE ABLATIONS

Each oracle condition (§7) replaces one stage of the trace with ground truth and leaves the rest of the pipeline unchanged. This isolates what that stage contributes as a diagnostic intervention, never a deployment setting. Read as a single aggregate, their effect is small and the analysis stops there. Read *per metric*, with metrics the intervention should not affect acting as a specificity control, the result is considerably stronger. Table 27 and Figure 17 give the per-metric effect of each oracle, on which the paragraphs below draw. Appendix L discusses them. **The conflict label moves behaviour and only behaviour.** Supplying the gold Stage-2 label improves Behaviour Adherence (+0.013, 8/9 cells, $p = 0.039$) (Table 27). **The document notes move grounding and only grounding.** Supplying gold Stage-1 notes improves every grounding and citation metric, each positive in *every* cell (Table 27). Behaviour shows a small spillover, expected since correct document verdicts also constrain the conflict inference. The answer/refuse decision is untouched, since GR is upstream of both interventions. **Interpretation.** That the effects add ($|\Delta_{\text{both}} - (\Delta_{\text{conflict}} + \Delta_{\text{notes}})| \leq 0.010$ on every metric, Appendix L) is itself a construct-validity argument for the decomposition. For *baseline* models every oracle instead *hurts*, catastrophically so for grounding under the conflict label (−0.238), a second internalization result. Appendix L gives the per-metric residuals and the baseline breakdown.

8.5 EXTERNAL PROMPTING COMPARATORS

We compare against two prompting-only strategies applied to baseline models, neither fine-tuned and both on the same benchmark and evaluator. These are few-shot Chain-of-Thought (Wei et al., 2022) and Chain-of-Note-style retrieval notes (Yu et al., 2024). Both reach usable answer content (Table 4), but neither acquires the answer/refuse decision. Chain-of-Note *never abstains* (Ref. F1 = 0 for every model, Table 4), landing exactly on the answer-everything floor. Few-shot Chain-of-Thought abstains inconsistently rather than rarely, from 32/128 to 85/128 across models. Structured prompting can therefore elicit conflict-aware *answers* without supplying conflict-aware *judgment*, the distinction §1 draws. Appendix I gives the per-model abstention spread and Chain-of-Note’s citation hygiene, the poorest of any method examined. **Citation behaviour.** Trace supervision changes citation *sufficiency*, not citation precision: precision is near-ceiling for every model regardless of tuning, while the recall side moves sharply. Appendix I gives the seven-metric analysis, the definitions and the per-cell values.

8.6 ANSWER-ONLY SUPERVISION COMPARISON

The comparators above are prompt-only. This one is a matched SFT comparator. We fine-tune Llama-3.1-8B and Qwen2.5-7B on the same 862 base examples and QLoRA configuration as the trace-supervised runs (which add 27–48 further examples, Appendix D), with the trace removed from the supervision target (Appendix E). Because base data, model and optimiser are otherwise held fixed, most of any remaining gap is attributable to the trace itself, not fully isolated from this volume difference. Under the minimal prompt, trace supervision exceeds this answer-only baseline on every metric for both models, with the largest gap on Refusal Precision (Table 16, Appendix E). Answer Quality (Appendix G.8), the geometric-mean combination of Factual Grounding and Single-Truth Recall used elsewhere in the paper, favours the trace as well. Appendix E gives the complete per-metric deltas and the recipe in full.

9 VALIDATING THE EVALUATOR

A judged metric is only as trustworthy as its judge, so we validated the CATS committee against an independent human study. Four reviewers double-judged 300 responses spanning both model families, both training conditions, and all five conflict types. Grounded Refusal is deterministic and therefore outside this exercise. Only the three judged components are at stake. **Factual Grounding and Single-Truth Recall are reliable** ($\kappa = 0.78$ and 0.79 against humans). **Behaviour Adherence is not** ($\kappa = 0.41$), with the weakness concentrated in Complementary information and Misinformation, where human reviewers also disagree most. We therefore report conflict-conditioned behaviour qualitatively and do not interpret individual BA margins. Appendices H and Q give the study design and coverage caveats. Any reading of a BA number should be checked against them.

10 CONCLUSION

Trace supervision transforms the answer/refuse decision and makes conflict-aware behaviour survive removal of the instruction describing it. It does *not* improve answer writing, because that deficit was never there to fix. The three stages behave as separable constructs, so the trace is a measurement instrument rather than a formatting convention. Three questions remain open. The residual error after both oracles is small but unattributed, and separating Stage-3 synthesis failure from annotation noise and judge strictness needs a targeted audit. Behaviour Adherence is the weakest instrument we report, so conflict-behaviour claims should stay qualitative until it improves. Misinformation remains the smallest class in the benchmark. Expanding it without perturbation-based synthesis is an open data-collection problem. The practical consequence is a separation of concerns. A retrieval system that answers well but cannot decide *when* to answer is not a weaker version of one that can. The two differ in a capability that no prompt in our ladder supplied, and that supervising the reasoning did. Building such a system means supervising the judgment, and evaluating it means scoring that judgment apart from the answer it produces.

LIMITATIONS

Per-document verdicts rest on a small, non-blind check. The Stage-1 verdicts that gate Factual Grounding eligibility come from the annotation committee. A targeted review of 100 query–document pairs (three reviewers, 300 ratings) found 92% agreement ($\kappa = 0.88$) with the human-majority verdict, against human–human agreement of $\kappa = 0.83$. Reviewers saw the committee verdict before rating, so anchoring may inflate this figure, and no independent gold-standard label exists.

CATS is evidence-bounded, not truth-bounded. It scores whether the retrieved set justifies an answer, not whether the answer is true. The grounding criterion is permissive by design, so our grounding numbers are an upper bound relative to strict entailment.

Behaviour Adherence is our least reliable instrument ($\kappa = 0.41$ against humans, $\alpha = 0.46$ agreement among judges). A priority-weighted vote rule makes disputed judgments closer to a single-judge call than the panel size suggests. We therefore report conflict-behaviour findings qualitatively rather than by margin. Appendix S gives six further limitations: model families, aggregate properties, recipe asymmetry, annotation quality, review coverage, and comparator scope.

REPRODUCIBILITY STATEMENT

We release the training/validation split, the held-out benchmark, the stagewise annotation prompts, the end-to-end inference prompts, the CATS judge prompts, and the per-run evaluation outputs from which every number in this paper is computed (Appendix N). Appendix D gives the QLoRA hyperparameters, the two training recipes, and the checkpoint-selection criterion for every fine-tuned model. Appendix F gives the deterministic decoding protocol (temperature 0, fixed continuation budgets, sentinel-based stopping) used for every generation reported in the paper. Appendix B gives the formal annotation aggregation rule and Appendix P gives the complete CATS judge prompts, so both the released labels and the evaluation metric can be independently reproduced. Appendix A documents the retrieval protocol, the source-balanced sampling rule, and the per-source composition of the benchmark.

ETHICS STATEMENT

Data provenance and licensing. We derive the released resource from publicly available research datasets and from live web retrieval. We preserve source provenance in every record. The dataset cards identify the upstream collection for each source family, so downstream users can consult the relevant licences before redistributing source-derived material. We store retrieved snippets with their source URL and, where available, a timestamp. Because live retrieval is time-varying, readers should treat the release as a snapshot, not a reproducible query against a current index.

Human annotators. Members of the research group, not crowdworkers, carried out the human review. We retain reviewer identifiers in the review artifacts for auditability, but they carry no demographic information. We make no claims about recruitment, compensation, or demographics that our records do not support.

Intended use and risks. We intend the resource for research on evidence-grounded generation, abstention calibration, and conflict-aware evaluation. First, conflict labels and answerability judgments describe the *supplied* evidence, not world truth. Readers must not treat them as factual adjudications of the underlying questions, several of which concern contested scientific or medical topics. Second, the source-quality field is a coarse URL-whitelist heuristic retained as annotation metadata. It is not a credibility ranking of publishers, and researchers should not use it as one.

AI USE STATEMENT

Required disclosure. We used frontier language models for *synthetic data generation*: they produced the synthetic annotations in our dataset and served as the *LLM-as-a-judge* evaluators of model performance; the methodology for both is specified in §4 and §6. No other required-disclosure task (developing the theoretical framework, formulating claims, designing the methodology or experiments, or implementing the fine-tuning and evaluation code) was performed by generative AI; these are the authors’ own.

Recommended disclosure. We also used generative AI tools to edit this manuscript. This included tightening prose, enforcing consistent table formatting and typography, verifying citation accuracy and cross-reference integrity, compressing the main text to the submission page limit, and drafting appendix analyses and disclosures added in response to internal review, with every number checked against the underlying released results before inclusion.

Accountability. All experimental design, analysis, and claims are the authors’ own. We reviewed every edited passage against the source results, and we take responsibility for the final content of this work, including text, claims, and artifacts produced with the aid of generative AI.

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APPENDIX

A RESOURCE CONSTRUCTION

A.1 BENCHMARK COMPOSITION BY SOURCE AND TRACK

Source	Track	n	NC	CI	CO	OI	MI
QACC	web-retr.	145	69	46	14	7	9
SituatedQA	web-retr.	141	43	50	6	32	10
ConflictingQA	web-retr.	137	8	53	73	1	2
TRUST-ALIGN	refusal	128	57	45	13	13	0
FreshQA	web-retr.	85	33	8	2	27	15
WikiRevision	curated	78	0	0	0	78	0
HotpotQA	curated	19	0	19	0	0	0
HealthContradict	curated	2	1	0	1	0	0
Misinformation	curated	1	0	0	0	0	1
Total		736	211	221	109	158	37

Table 5: **Benchmark composition by source and construction track.** NC = No conflict, CI = Complementary information, CO = Conflicting opinions/research outcomes, OI = Conflict due to outdated information, MI = Conflict due to misinformation. Track A is web-retrieved, Track B is the held-out refusal pool, Track C is curated conflict. The curated track is class-targeted: all 78 WikiRevision items are temporal-supersession cases, supplying 49% of that class, and all 19 HotpotQA items are complementary-evidence cases.

Table 5 gives the full breakdown referenced in §4. The three tracks are structurally distinguishable, which is why we report them separately rather than as one pool. Track A records carry live retrieval URLs across 1,423 distinct domains, and 503 of the 508 records have exactly five documents each

Stage	Records
Raw candidate rows from four source pools	20,330
— normalised overlaps with the training question set	−598
— duplicate normalised queries	−8,891
Unique usable candidates	10,841
Source-balanced selection (seeded)	2,000
Retained with exactly ten documents	1,878 (93.9%)
underfilled, excluded	122
Human first-pass reviewed (canonical cleaned corpus)	1,221
Selected non-refusal set	800
Consolidated preselection/review population	1,454
Final answerable benchmark records	608

Table 6: Non-refusal construction funnel, counts as observed in the retained artifacts.

(5 records have six to eight). Track B records use placeholder URLs inherited from the upstream refusal resource and are uniformly five documents. Track C records are internally homogeneous by source: all 78 Wikipedia-revision items have exactly four documents drawn from a single domain with revision-dated URLs. All 19 multi-hop items have exactly ten documents with no URLs.

A.2 TRAINING SET COMPOSITION

The training release combines 458 normalised examples from the CONFLICTS collection (Cattan et al., 2025) and 272 held-out refusal examples derived from Trust-Align (Song et al., 2025). It also includes 213 examples drawn from the benchmark construction stream that were not admitted to the final benchmark. All 213 were checked for identifier overlap against the released benchmark. The intersection is empty.

A.3 RETRIEVAL FUNNEL

The non-refusal track proceeds through the following stages (Table 6). Counts are as observed in the retained artifacts.

Retrieval requests twenty results per query and retains at most ten, with a query-focused window selected by a TAS-B dense bi-encoder (Hofstätter et al., 2021). The released records carry a deterministic five-document view of those ten, sampled by a seeded rule that takes two from the top five and three from the bottom five. This is why Table 1 reports a mean of 5.03 documents per benchmark record rather than ten. The reduction keeps the evidence set answerable. It also prevents the conflicting documents from always occupying the highest-ranked positions, which would otherwise let a model succeed by attending to rank alone.

Two discontinuities in this funnel have no retained per-record selector and we report them as observed rather than reconstructing a rule. First, 1,878 candidates were assigned for first-pass review but the canonical cleaned corpus contains 1,221 unique reviewed records. The remaining discontinuity is that the quality selection from the 1,454-record consolidated population down to the released benchmark is described by a documented predicate, acceptance, reviewer confidence, retrieval quality, evidence sufficiency, conflict clarity, query specificity, source reliability, relevant-document count and gold-answer feasibility. No per-record script for that selection was retained. Relatedly, 105 of the 608 answerable benchmark records appear in the consolidated review population only as consensus-completed entries rather than as first-pass reviews. The refusal track reduces from a 200-record curated subset to the 128 released records without a documented reduction rule. A 128-record refusal-quality review stratum with complete agreement is retained, however.

A.4 VERIFICATION OF THE RELEASED FILES

All statistics below are computed directly from the released JSONL rather than from construction logs (Table 7; Figures 3 and 4).

Property	Train	Benchmark
Records	943	736
Duplicate identifiers	0	0
Identifier overlap (train/val $\cap$ benchmark)	0	0
Document/note identifier misalignments	0	0
Canonical d1 . . dN ordering	943/943	736/736
Key fact on every non-irrelevant note	5,886/5,886	3,393/3,393
Irrelevant notes carrying a key fact	0	0
Notes retaining an annotator vote tally	6,642/6,642	3,701/3,701
Unique source domains	2,633	1,431
Mean snippet length (words)	231	222
Quote is an exact contiguous span	82.6%	81.1%
Quote is snippet-anchored (exact/ellipsis/partial)	95.6%	95.9%

Table 7: Verification of the released files, computed directly from the released JSONL rather than from construction logs.

An independent validity check on the refusal track. Refusal-required examples are not merely *labelled* insufficient. The document-level adjudication, which was performed independently of the abstention label, confirms it. Decisive *supports* verdicts occur on 3.6% of refusal-track documents against 39.3% elsewhere, an eleven-fold difference.

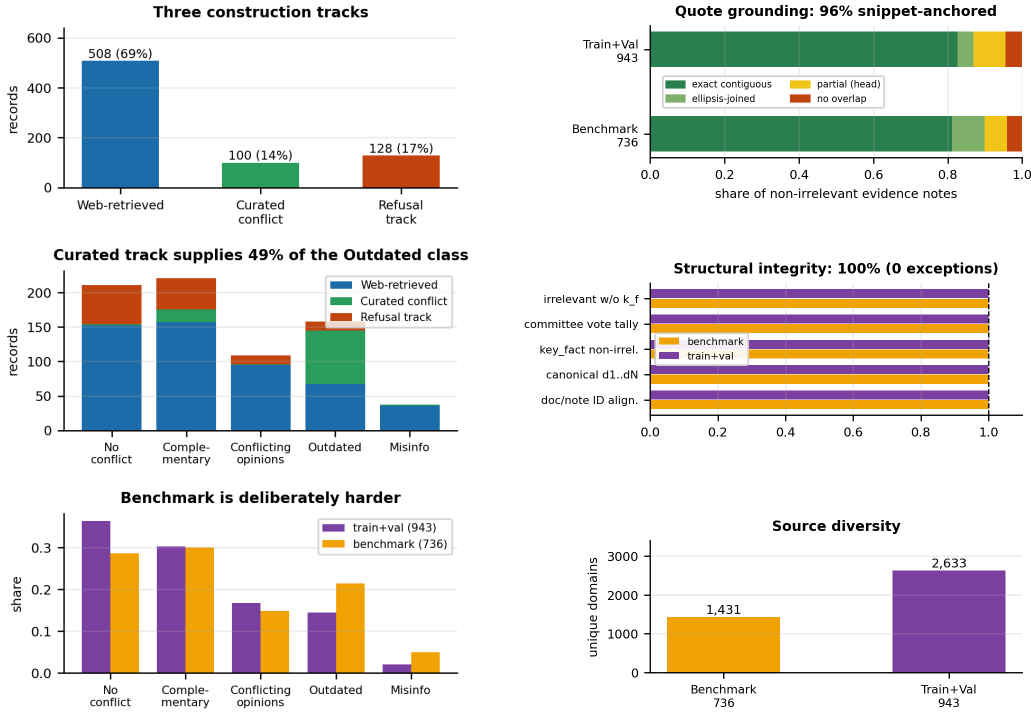


Figure 3: Resource composition. Left: the three benchmark construction tracks. Centre: the curated track is class-targeted, supplying 49% of the temporal-supersession class. Right: the benchmark is deliberately harder than the training distribution, with fewer no-conflict and more outdated/misinformation cases; **Figure 4:** Quality audit computed directly from the released files.

B ANNOTATION PIPELINE

Annotation is stagewise. At each stage every committee member independently receives the same structured input and returns a structured output. The committee then votes only on the central

categorical decision for that stage and adopts the associated natural-language explanation *intact* from the highest-weight member that supported the winning decision. The method therefore combines committee agreement on decisions with a single internally coherent reasoning bundle, and it never averages or splices prose from different annotators.

Formally, for a decision field with candidate values v , each successfully represented annotator m contributes its configured weight w_m and the system selects

$$v^* = \arg \max_v \sum_m w_m \mathbf{1}[y_m = v].$$

Exact ties are broken deterministically by a fixed candidate ordering, so that re-running the pipeline on identical inputs reproduces identical labels. Writing $\mathcal{M}^* = \{m : y_m = v^*\}$ for the annotators on the winning side, the explanation bundle is adopted *intact* from $m^* = \arg \max_{m \in \mathcal{M}^*} w_m$, rather than assembled field-by-field across annotators. As a result, a verdict, its quote and its rationale always originate from one coherent interpretation. Every note retains `_vote_tally`, `_winner_model` and `_all_verdicts`, so any label can be traced to its vote.

Stage-1 output is constrained. It always includes a verdict from the closed set, a key fact anchored to a quoted span of at most 50 words, a verdict rationale and a URL-derived source-quality label. The benchmark-mode Stage-1 prompt differs from the ordinary prompt in one consequential way. Older, contradictory or otherwise conflict-bearing evidence about the target is retained as *partially supports* rather than discarded as *irrelevant*. This lets Stage 2 see the evidence needed to diagnose temporal and scientific conflicts at all. Stage-2 votes the conflict label and the answerability flag. Stage-3 votes the abstention decision independently of Stage-2 answerability. This is why a complete analysis must inspect both fields.

Different retained artifacts were produced by different committee configurations and we report the configuration alongside the artifact rather than applying a single current default retrospectively. Structural validators check schema conformance, label inventories and document/note alignment. They do not establish quote entailment, factual correctness or temporal ordering. This is why the human review of Appendix C is reported separately rather than as a property of the automated pipeline.

C HUMAN REVIEW OF THE RESOURCE

Three workflows cover three different review targets and must not be conflated. All are released with their raw per-reviewer files, assignment manifests, consolidated pair records and machine-readable agreement metrics.

C.1 STAGE-1 DOCUMENT-VERDICT REVIEW

The reviews below check the *conflict-type* label and the *preselection* decision. Neither directly checks the Stage-1 per-document verdict (*supports*, *partially supports*, *irrelevant*) that gates Factual Grounding eligibility (§3). A separate, smaller study targets that verdict directly.

Design. We drew a sample of 100 query–document pairs with existing Stage-1 LLM-committee verdicts, seeded (20260919). The sample is balanced by committee verdict (supports: 34; partially supports: 33; irrelevant: 33) and deduplicated so each normalised query appears once. The pairs are drawn from the holdout benchmark (46), the training set (50) and the validation set (4). Three reviewers each assessed all 100 pairs (300 ratings). Each saw the query, the document, and the committee’s Stage-1 annotation, and recorded agreement or a replacement verdict with a reason. Because reviewers saw the committee verdict before recording their own, the review is not blind, and anchoring may inflate the agreement figures below. There is also no independent gold-standard label, so these are agreement statistics, not accuracy statistics.

Human–human agreement. Table 8 gives pairwise and three-rater agreement. Three-way unanimity holds on 83/100 pairs, with a 2–1 majority on the remaining 17 and no three-way splits.

Committee–human agreement. Table 9 compares the committee’s Stage-1 verdict against each reviewer and against the human majority. The pooled row treats each of the 300 ratings as a paired

Comparison	Pairs/ratings	Exact agreement	κ
Reviewer 1 vs Reviewer 2	100	88.0%	0.820
Reviewer 1 vs Reviewer 3	100	85.0%	0.776
Reviewer 2 vs Reviewer 3	100	93.0%	0.895
Fleiss' κ (3 raters)	100	—	0.830
Krippendorff's α (nominal)	300	—	0.830
Mean pairwise exact agreement	300	88.7%	—
Unanimous verdicts (3/3)	100	83.0%	—
2–1 majority (no 3-way ties)	100	17.0%	—

Table 8: **Stage-1 review: human–human agreement** across the three reviewers, 100 pairs (300 ratings). “—” marks a cell the statistic does not apply to.

observation against the same committee label repeated per reviewer. It is descriptive, since the ratings cluster within 100 pairs.

Comparison	Pairs/ratings	Exact agreement	κ
Committee vs Reviewer 1	100	96.0%	0.940
Committee vs Reviewer 2	100	90.0%	0.850
Committee vs Reviewer 3	100	87.0%	0.805
Committee vs human majority	100	92.0%	0.880
Committee vs all individual ratings*	300	91.0%	0.865

Table 9: **Stage-1 review: committee–human agreement.** * Pooled statistic. The 300 ratings share 100 underlying pairs.

Where the committee and reviewers disagree. Table 10 gives the pooled confusion matrix against all 300 individual ratings. It also gives the correction directions among the 27 ratings (18 distinct pairs) where a reviewer replaced the committee’s verdict. Every recorded correction is between adjacent categories: no reviewer moved a pair directly between *supports* and *irrelevant*, matching the pattern in the human–human disagreements (*supports* vs *partially supports* on 11 pairs, *partially supports* vs *irrelevant* on 6, no *supports*–*irrelevant* disagreement). Agreement is not uniform across the committee’s own verdict: conditional on the committee saying *irrelevant*, reviewers agree on 94.9% of ratings (94/99); conditional on *supports*, 89.2% (91/102); conditional on *partially supports*, 88.9% (88/99). The committee is therefore most reliable exactly where Factual Grounding eligibility is coarsest, on whether a document is usable at all, and least reliable on the finer *supports*-versus-*partially supports* distinction that does not affect eligibility.

Committee verdict	Human: supports	Human: partially	Human: irrelevant	Total
Supports	91	11	0	102
Partially supports	5	88	6	99
Irrelevant	0	5	94	99

Committee to human verdict	Correction ratings	Distinct pairs
Supports to partially supports	11	7
Partially supports to supports	5	4
Partially supports to irrelevant	6	4
Irrelevant to partially supports	5	3

Table 10: **Stage-1 review: pooled confusion matrix** (top, 300 ratings against the repeated committee label) **and correction directions** (bottom, among the 27 ratings that changed the committee’s verdict).

Interpretation and limitations. Reviewers agree with one another and with the Stage-1 committee label on this sample, and most committee–human differences sit at neighbouring category boundaries. The sample is deliberately balanced by committee verdict, so it does not estimate agreement at the datasets’ natural label prevalence, and it mixes holdout, training and validation records, so it must not be read as a holdout-only evaluation. Read together with the non-blind design, this study supports using the committee’s Stage-1 verdict as a useful first-pass signal, checked against human judgment on this one sample. It does not establish the verdict’s objective accuracy across the corpus (§10).

C.2 TRAINING CONFLICT-TYPE REVIEW

All 658 base records were assigned to exactly two of three reviewers under a seeded, pair-balanced, label-interleaved allocation. This gives 1,316 assignments with loads of 439/439/438 and reviewer-pair counts of 220/219/219. Reviewers saw the query, the retrieved snippets and the committee-assigned label, and could retain it or replace it, recording confidence and a reason. This is a label-validation protocol, not a keep/drop filter: no item is removed from the pool by review.

Review field	agree	raw	κ	Review field	agree	raw	κ
				Preliminary conflict type	1378	94.77%	0.9217
				Retention decision	1428	98.21%	0.9228
Reviewed conflict type	787	83.46%	0.7694	Retrieval quality	1436	98.76%	0.9550
				Evidence sufficiency	1436	98.76%	0.9600
Retain-or-change action	791	83.88%	0.1341	Conflict clarity	1441	99.11%	0.9660
				Query specificity	1449	99.66%	0.9875
Changed-label (boolean)	791	83.88%	0.1341	Source reliability	1446	99.45%	0.9881
				Relevant-document count	1426	98.07%	0.9527
Reviewer confidence	607	64.37%	0.0159	Gold-answer feasibility	1440	99.04%	0.9669
				Reviewer confidence	1445	99.38%	0.9767

Table 11: Training conflict-type review agreement over the 943-record consolidated population. The five-way taxonomy field is the primary result. The low chance-corrected values on the remaining rows reflect strongly imbalanced marginals (most reviewers accepted the committee label and reported high confidence), not label-quality failures: raw agreement and κ answer different questions when one category dominates; **Table 12: Benchmark preselection agreement**, all ten recorded review fields over the 1,454-record consolidated population. Every field exceeds $\kappa = 0.92$. The primary result is the five-way conflict-type field. Because the second pass is sequential (Appendix C), these figures measure verification consistency rather than blind double annotation.

The primary result in Table 11 is the five-way conflict-type field: 787 of 943 pairs agree, giving 83.46% raw agreement and $\kappa = 0.7694$.

What the review layer actually changed. Relative to the original committee label, the first review retains it for 801/943 records (84.94%), the second for 895/943 (94.91%), and both reviews retain it for 772/943 (81.87%). The review layer is therefore doing real work: roughly one record in five is contested by at least one reviewer.

Reviewer labels and released labels are different quantities. The consolidation deliberately does not overwrite the released `conflict_type`. The two distributions consequently differ, and we report both rather than implying the review ratified the release. Reviewers’ first-pass labels give No conflict 373, Complementary 289, Outdated 130, Conflicting opinions 120 and Misinformation 31, against released counts of 343, 286, 136, 158 and 20. The largest divergences are the conflicting-opinions and misinformation classes. Any analysis that treats the released label as human-ratified would therefore overstate the evidence. We treat the review as a reliability estimate on the taxonomy, not as a corrected gold standard.

C.3 BENCHMARK PRESELECTION REVIEW

Seven reviewers were allocated the 1,878 exact-ten candidate set under seeded source balancing, with non-uniform queues of 150/150/300/300/326/326/326. The source mix is ConflictingQA 258, FreshQA 394, SituatedQA-temporal 413, QACC 407 and SituatedQA-geographic 406. Each

reviewer recorded an acceptance decision, a preliminary conflict type, confidence, retrieval quality, evidence sufficiency and conflict clarity. They also recorded query specificity, source reliability, a relevant-document count bin, gold-answer feasibility and free-text notes.

Every one of the ten recorded fields in Table 12 exceeds $\kappa = 0.92$, with the five-way conflict-type field at 94.77% raw agreement and $\kappa = 0.9217$ (1,378 of 1,454 pairs agreeing).

The second pass is sequential. The second reviewer sees the complete first review and may accept, edit or reject it. Across the 800 selected records there are 673 accept-as-is and 127 edited actions. These figures therefore measure verification consistency and must not be presented as blind independent double annotation. The second pass was nonetheless able to dissent. It introduced label values the first pass never used, including 13 *Unsure* conflict-type judgments and 15 reject or borderline-reject retention decisions.

Reconstruction caveat. Three second-review output files were lost. They were reconstructed after the corresponding reviewers confirmed that every first review in their queue had been accepted as correct (114, 115 and 114 records). Those rows are therefore attested rather than independently recorded, and the reconstruction manifest is released alongside the review data.

C.4 REFUSAL-QUALITY VALIDATION

The 128 released refusal items were reviewed as a separate stratum against four dedicated fields in addition to the ten preselection fields. Agreement is *unanimous on all fourteen*: both reviewers judged all 128 items to require refusal, to have a valid abstention ground truth, to carry a high-quality evidence-gap rationale, and to be a valid refusal benchmark item. They also agreed unanimously that evidence is insufficient and that no defensible gold answer is possible for any of them. Because each of these fields is single-valued across the stratum, κ is undefined and we report exact agreement rather than substituting a coefficient.

This complements the automated check in Appendix A: decisive *supports* verdicts occur on 3.6% of refusal-track documents against 39.3% elsewhere. The refusal targets are thus corroborated twice, once by document-level adjudication performed independently of the abstention label and once by unanimous human review.

D FINE-TUNING: RECIPE CHRONOLOGY

D.1 OPTIMISATION AND CHECKPOINT SELECTION

All models are fine-tuned with QLoRA (Detrmers et al., 2023; Hu et al., 2021) at rank 32, $\alpha = 64$ and dropout 0.05, with learning rate 2×10^{-4} , effective batch 16, maximum sequence length 12,288 and NEFTune (training-time embedding noise) $\alpha = 5.0$ (Jain et al., 2023). Training minimises the standard token-level cross-entropy of the serialised trace and final response given the prompt and retrieved documents. No auxiliary loss term is added. Checkpoints are selected on a held-out validation split by a composite score that sums document-verdict accuracy, response-format validity and abstention correctness, penalising false abstentions so that a model cannot score well by refusing answerable questions. A *recipe* is one fixed training mixture: the set and proportion of task views and prompt variants a run is trained on. Qwen2.5-7B and Qwen2.5-32B use recipe K. Llama-3.1-8B and Mistral-7B use recipe L.

The recipe was developed by error-directed revision rather than search. Table 13 names each intervention and its measured outcome, including the two steps that did not work.

Here “n/r” marks a step whose exact mixture size was not retained. Row counts are repeated supervised *views*, not independent questions: recipe L draws on 910 source examples. Three cautions apply to any causal reading. J, K and L each change data coverage, drill composition and sample weighting together, so they are principled recipe revisions rather than single-factor ablations. The source-hygiene intervention explored at step G is *not* present in the K/L mixtures and should not be attributed to the reported models. And the reported matrices use recipe K for the Qwen family and recipe L for Llama and Mistral, because no completed Qwen matrix exists for recipe L.

Step	Rows	Intervention and outcome
D	6,699	Strict/runtime/minimal multi-task mixture. <i>Established that a trace survives minimal prompting.</i>
E	n/r	Broad conflict-label pressure. Negative: improved 32B conflict accuracy, degraded 7B. Capacity-sensitive.
F	7,308	One targeted taxonomy-boundary drill per source example. Best historical conflict result at both scales; narrower and more effective than E.
G	8,526 / 8,715	Model-specific probes: extra Stage-1 supervision (32B) and source-hygiene plus minimal-prompt pressure (7B). Negative: each recovered its target metric but cost conflict/contract quality.
J	10,674	<i>Turning point.</i> Rebuilt train/validation geometry to include benchmark-like short answerable cases. Broad over-abstention substantially reduced; residual errors became localised.
K	12,659	+27 derived five-document answerable examples, document-boundary and partial-synthesis drills. Clean gain for 32B; genuine trade-off for 7B.
L	13,349	+21 short answerable <i>no-conflict</i> examples, correcting a shortcut K risked introducing. Strong for Llama; Mistral still over-abstains.

Table 13: **Fine-tuning recipe chronology.** Row counts are repeated supervised views, not independent questions. “n/r” marks a step whose exact mixture size was not retained.

Benchmark awareness during recipe development. Step J’s training augmentation was constructed by sampling additional answerable examples from the same source pool that the released 736-example holdout was later drawn from: of 741 benchmark-pool rows judged answerable, 193 were selected into that step’s augmented training/validation split (168 train, 25 validation) and the remaining 548 became part of the residual pool the released holdout is drawn from. The final released training set and the released 736-example holdout share zero identifiers and zero normalised-query matches, which we verified directly on the released files. This identifier-level disjointness is a narrower guarantee than full benchmark blindness, however: archived evaluation records from this period show the 736-example holdout being scored during recipe iteration, so recipe choices at this stage were made with some visibility into holdout performance and into the benchmark pool’s general characteristics, not with the holdout withheld in the stronger sense of never having been consulted at all. We report this precisely rather than claim a blindness the development process did not have.

E ANSWER-ONLY SUPERVISION ABLATION

E.1 RECIPE

This comparator isolates what the evidence trace itself contributes, holding most of the data and the optimisation fixed. It reuses the same 862/81/736 train/validation/benchmark split and the identical QLoRA configuration (rank 32, $\alpha = 64$, dropout 0.05, learning rate 2×10^{-4} , maximum sequence length 12,288, NEFTune $\alpha = 5.0$) as the trace-supervised runs, changing the supervision target. It is not, however, trained on exactly the same source examples as those runs: the trace-supervised recipes augment the base 862 with 27 further derived examples for recipe K (Qwen) and 48 for recipe L (Llama, Mistral), so the trace-supervised side of this comparison has slightly more source data, not just a different supervision target (Appendix D). The assistant message contains *only* the final grounded answer or the canonical refusal string, immediately followed by [[END-OF-ANSWER]]. It contains no <think> block, document-verdict array or conflict-type label. Two prompt families are trained: **final-only**, which explicitly instructs the model to omit the trace and produce only the final answer, and **minimal**, the same forty-word evidence policy used throughout the paper. Each of the 862 source examples yields 8 final-only and 4 minimal copies (a 2:1 mixture), giving 10,344 training rows. Generation is deterministic (temperature 0), with the same 4-bit base-model loading and LoRA adapter as the trace-supervised runs. Evaluation runs the two prompt families through refusal-decision metrics: precision, recall and F1 for the answer and the refuse class, and overall classification accuracy. On the same generations, it also computes the paper’s own Behaviour Adherence, Factual Grounding and Single-Truth Recall components. Answer Quality is the same AQ quantity defined in Appendix G.8 (the geometric mean of Factual Grounding and Single-Truth Recall, or Factual Grounding alone where Single-Truth Recall does not apply). It is reported for the

trace-supervised runs in Table 27 and Table 33, and computed here for the answer-only checkpoints as well.

This ablation covers Llama-3.1-8B and Qwen2.5-7B only, not the full four-model matrix used elsewhere in the paper. It also isolates the whole trace, all three stages together, from no trace at all; we did not train a model with only a subset of the three stages supervised (for example Stage-1 verdicts alone, or Stages 1 and 2 without Stage 3), so the marginal contribution of each individual stage to this specific gap remains an open question for future work. This is distinct from the oracle ablations of §8.4, which inject gold information at inference time rather than varying what is supervised during training.

E.2 RESULTS

Tables 14 and 15 give the complete ladder: both answer-only prompt variants against all three trace-supervised prompt variants, per model. Table 16 isolates the minimal-prompt comparison, the condition §8.2 identifies as the most diagnostic, since it carries the least prompt scaffolding for either supervision type to lean on.

Metric (%)	Answer-only		Trace		
	minimal	final-only	minimal	runtime	strict
Answering Precision	98.41	98.40	99.83	99.83	99.66
Answering Recall	91.45	91.28	97.20	97.04	97.04
Answering F1	94.80	94.71	98.50	98.42	98.33
Refusal Precision	69.59	69.19	88.19	87.59	87.50
Refusal Recall	92.97	92.97	99.22	99.22	98.44
Refusal F1	79.60	79.33	93.38	93.04	92.65
Overall Classification Accuracy	91.71	91.58	97.55	97.42	97.28
Behavioural Adherence	72.11	71.10	79.47	80.62	80.49
Answer Quality	57.80	54.90	69.40	69.00	70.60
Single-Truth Recall	69.64	68.17	71.36	70.53	72.69
Factual Grounding	71.83	70.47	85.74	84.91	85.16

Table 14: **Llama-3.1-8B: answer-only vs. trace supervision**, all five prompt×supervision cells. Values are percentages, as reported by the evaluation harness.

Metric (%)	Answer-only		Trace		
	minimal	final-only	minimal	runtime	strict
Answering Precision	96.35	97.37	97.84	98.83	99.16
Answering Recall	91.28	91.28	97.04	97.04	96.88
Answering F1	93.75	94.23	97.44	97.93	98.00
Refusal Precision	66.88	68.07	86.47	87.05	86.62
Refusal Recall	83.59	88.28	89.84	94.53	96.09
Refusal F1	74.31	76.87	88.12	90.64	91.11
Overall Classification Accuracy	89.95	90.76	95.79	96.60	96.74
Behavioural Adherence	73.18	73.19	76.81	77.40	77.98
Answer Quality	51.20	53.50	68.00	69.00	67.70
Single-Truth Recall	61.27	62.20	68.58	69.30	68.69
Factual Grounding	68.34	70.24	83.75	82.91	83.84

Table 15: **Qwen2.5-7B: answer-only vs. trace supervision**, all five prompt×supervision cells. Values are percentages, as reported by the evaluation harness.

Under the minimal prompt, trace supervision exceeds answer-only supervision on every metric for both models: Refusal F1 by +13.78pp (Llama-3.1-8B) and +13.82pp (Qwen2.5-7B), Factual Grounding by +13.92pp and +15.41pp, and Answer Quality by +11.60pp and +16.80pp. Refusal Precision shows the largest single gap (+18.60pp and +19.59pp): answer-only supervision, on

Metric (%), minimal prompt	Llama-3.1-8B			Qwen2.5-7B		
	AO	Trace	$\Delta(\text{pp})$	AO	Trace	$\Delta(\text{pp})$
Answering Precision	98.41	99.83	+1.42	96.35	97.84	+1.49
Answering Recall	91.45	97.20	+5.76	91.28	97.04	+5.76
Answering F1	94.80	98.50	+3.70	93.75	97.44	+3.69
Refusal Precision	69.59	88.19	+18.60	66.88	86.47	+19.59
Refusal Recall	92.97	99.22	+6.25	83.59	89.84	+6.25
Refusal F1	79.60	93.38	+13.78	74.31	88.12	+13.82
Overall Classification Accuracy	91.71	97.55	+5.84	89.95	95.79	+5.84
Behavioural Adherence	72.11	79.47	+7.36	73.18	76.81	+3.64
Answer Quality	57.80	69.40	+11.60	51.20	68.00	+16.80
Single-Truth Recall	69.64	71.36	+1.72	61.27	68.58	+7.31
Factual Grounding	71.83	85.74	+13.92	68.34	83.75	+15.41

Table 16: **Answer-only vs. trace supervision under the minimal prompt**, both models, with the percentage-point (pp) difference. AO is answer-only supervision.

identical data and compute, does not learn the same calibrated answer/refuse decision that the trace does. Behaviour Adherence and Single-Truth Recall also favour the trace in every cell, though by a smaller and less even margin (+7.36pp/+3.64pp and +1.72pp/+7.31pp respectively). This is a matched comparison: because the two supervision types share training data, base model and optimiser, the gap is attributable to the presence of the evidence trace in the supervision target, not to additional data or compute.

F INFERENCE PROTOCOL

Generation is deterministic at temperature 0. Continuation budgets are profile-specific, strict 1,400 tokens base with a 3,200 cap, runtime 1,200/2,200, minimal 900/1,800, because a strict trace must enumerate every retrieved document and a uniform cap would selectively truncate long-context examples, confounding reasoning quality with output-space availability. Generation halts on the sentinel.

A retry re-runs the *identical* deterministic prompt with a larger token budget when the structural contract is unmet. It never resamples, appends corrective feedback, alters the retrieved documents or substitutes a prompt profile. The minimal profile has no structural retry gate by default, specifically to prevent the generation wrapper from turning a sparse-prompt condition into a hidden strict condition.

Every run writes both a raw and a sanitised output file. Sanitisation may canonicalise document order and identifiers, normalise label aliases, clear key facts on irrelevant documents, drop out-of-range citations and append a missing sentinel where a usable trace exists. It *cannot* invent a missing trace, construct absent document judgments, write a conflict analysis or supply a missing answer. A record without a parseable trace passes through unchanged.

F.1 OUTPUT LENGTH AS A COST PROXY

Emitting the structured trace has a real generation cost: the serialised `<think>` block adds Stage-1 per-document verdicts, a Stage-2 conflict label and a Stage-3 synthesis record before the final answer, none of which a direct-answer or prompting-only baseline produces. We measure this cost the way it is actually recoverable from the released generations: character length of the raw saved output (not a tokeniser count) over the full 736-example benchmark, reported as mean / median / 95th-percentile (nearest-rank).

Table 17 gives trace-tuned, answer-only-supervised and direct-answer baseline lengths for all four models; Table 18 gives the two prompting-only comparators for the two models whose raw generations were retained. Trace-tuned outputs run roughly $10\times$ longer than the corresponding answer-only outputs for Qwen2.5-7B and Llama-3.1-8B, and $4.6\text{--}8.6\times$ longer than the direct-answer baseline, depending on the model (smallest for Qwen2.5-32B, largest for Llama-3.1-8B). Mistral-7B’s answer-

Model	Trace-tuned SFT			Answer-only SFT			Direct-answer baseline		
	mean	med.	p95	mean	med.	p95	mean	med.	p95
Qwen2.5-7B	2495.1	2376.5	3557	253.9	88.5	838	377.2	242.5	1164
Llama-3.1-8B	2424.9	2290.0	3434	240.5	91.0	802	280.9	91.0	1359
Mistral-7B	2575.1	2431.5	3584	1545.0	1508.0	1924	302.1	202.5	800
Qwen2.5-32B	2421.8	2302.0	3445	–	–	–	525.8	410.5	1469

Table 17: **Output length by supervision condition**, characters over the 736-example benchmark under the minimal prompt. Qwen2.5-32B’s answer-only raw generations were not retained.

Model	Chain-of-Note			Few-shot Chain-of-Thought		
	mean	med.	p95	mean	med.	p95
Qwen2.5-7B	1827.7	1766.0	2637	2092.9	1960.0	3160
Mistral-7B	1661.1	1579.0	2461	2144.2	2044.0	3411

Table 18: **Output length for the two prompting-only comparators**, characters over the 736-example benchmark. Llama-3.1-8B and Qwen2.5-32B are omitted: only reformatted, not raw, generations were retained for their prompting-only runs.

only model is the one exception to the first comparison, itself emitting long, verbose answers (1,545 characters on average) despite receiving no trace supervision.

We do not report a matched wall-clock or tokens-per-second comparison. Generation timing was logged only as an aggregate per-job duration on a shared cluster, not as isolated per-example decoding latency, and the available job records mix GPU types across conditions in a way that is not directly comparable. A rigorous latency comparison would need generation re-run under identical, controlled hardware and batching, which we leave to future work; the character-length figures above are the load-bearing evidence for the trace’s generation-cost overhead in this paper.

G CATS SPECIFICATION

G.1 COMPONENT DEFINITIONS

Grounded Refusal (GR). Whether the system answered exactly when the evidence permits answering. It is computed *deterministically* from the benchmark answerability label, never by a judge. We report refusal F1, where a true positive is the model declining to answer when the retrieved evidence does not support one.

Behaviour Adherence (BA). Whether the answer followed the response policy its conflict type requires, judged against a per-type rubric. A no-conflict item expects a direct answer. Complementary evidence expects synthesis across documents. Conflicting opinions expect both positions represented without adjudication. Outdated information expects the current value with the superseded one flagged, and misinformation expects explicit correction. The rubric is reproduced verbatim in Appendix P.

Factual Grounding (FG). Whether each extracted claim is both supported by an eligible retrieved document *and* cited. Only documents whose *gold* per-document verdict is *supports* or *partially supports* are eligible. As a result, an answer that is true in the world but unsupported by the retrieved set scores zero by construction. Support may be distributed across a document pair.

Single-Truth Recall (STR). Whether the response explicitly adopts the target answer as its own conclusion. It fails this test if the response merely quotes the answer, presents it as one of several possibilities, or attributes it to a source without endorsing it. It is evaluated only for items where a single correct conclusion exists: no conflict, complementary information, outdated information and misinformation ($n = 487$). Conflicting-opinion items are excluded because no such conclusion exists for them.

Conflict type	Definition	Expected behaviour
No conflict	All sources provide consistent and aligned information.	Return a unified, concise answer that synthesises the content accurately.
Complementary information	Sources provide different but non-contradictory pieces that complete each other.	Integrate all valid details into a richer, more comprehensive response.
Conflicting opinions or research outcomes	Sources disagree due to different interpretations.	Present each viewpoint clearly, describe the nature of the disagreement, and avoid taking a side unless the evidence strongly supports one.
Outdated information	Some sources contain older data or conclusions that may no longer be reliable.	Prioritise newer, verified information while explicitly noting which sources are outdated.
Misinformation	A source provides factually correct, misleading or fabricated facts, and briefly explain why the claim is wrong.	Reject or correct the misinformation, give the verified claims.

Table 19: **The five conflict types**, their definitions and the response behaviour each requires. Stage 2 assigns exactly one label per example; Behaviour Adherence (§6) scores the response against the expected behaviour for the assigned type.

G.2 APPLICABILITY AND COMPLETENESS

For example i , let $A_i \in \{0, 1\}$ be gold answerability, $\hat{A}_i$ the parsed answer/refuse decision and $t_i \in \{1, \dots, 5\}$ the conflict type. For a metric M , write $\mathcal{I}_M$ for its applicable examples and $n_M = |\mathcal{I}_M|$. Every dataset-level component below is an example-macro average over that set, never an average over placeholder zeros. Correct refusals carry no answer-content metrics, so

$$\mathcal{I}_{BA} = \mathcal{I}_{FG} = \{i : \neg(A_i = 0 \wedge \hat{A}_i = 0)\},$$

$$\mathcal{I}_{AQ} = \mathcal{I}_{FG},$$

and $\mathcal{I}_{STR} \subseteq \mathcal{I}_{FG}$ additionally requires a gold answer and a single-truth conflict type. An *incorrect* refusal remains in these denominators with zero extracted content. It is not silently excluded. A headline aggregate is published only when every applicable example has a score and every represented type has a value. Otherwise the run is marked incomplete rather than silently changing the denominator.

G.3 GROUNDED REFUSAL

The per-example decision gate is $g_i = 1[\hat{A}_i = A_i]$. With “answered” as the positive class,

$$TP = \sum_i 1[\hat{A}_i=1, A_i=1],$$

$$FP = \sum_i 1[\hat{A}_i=1, A_i=0],$$

$$FN = \sum_i 1[\hat{A}_i=0, A_i=1],$$

$$TN = \sum_i 1[\hat{A}_i=0, A_i=0],$$

giving $P_{ans} = TP/(TP + FP)$, $R_{ans} = TP/(TP + FN)$ and $F_{1,ans} = 2P_{ans}R_{ans}/(P_{ans} + R_{ans})$, with a zero denominator yielding zero. The refusal-positive orientation swaps $(TP, FP, FN, TN) \rightarrow (TN, FN, FP, TP)$:

$$P_{ref} = \frac{TN}{TN + FN}, \quad R_{ref} = \frac{TN}{TN + FP},$$

and analogously $F_{1,ref}$. Decision accuracy is $(TP + TN)/N$. Both orientations are reported because a model can have near-perfect answer recall while refusing far too rarely. They are also reported because the benchmark is 82.6% answerable, so answering everything already yields $608/736 = 0.826$ accuracy (§8.1).

G.4 COMMITTEE VOTING RULE FOR BA AND STR

The active local committee is Qwen3.5-397B-A17B, Mistral Small 4 and DeepSeek-R1-Distill-32B with priorities $p_j = (6, 3, 2)$. On a binary judgment, valid judge j emits vote $z_{ij} \in \{0, 1\}$ and confidence $c_{ij} \in [0, 1]$, contributing effective weight and vote masses

$$a_{ij} = p_j \max(c_{ij}, 0.01),$$

$$W_i^+ = \sum_{j: z_{ij}=1} a_{ij}, \quad W_i^- = \sum_{j: z_{ij}=0} a_{ij}.$$

The decision and the continuous positive support are

$$d_i = \mathbf{1}[W_i^+ > W_i^-], \quad u_i = \frac{W_i^+}{W_i^+ + W_i^-}.$$

A weighted tie is negative. Timeout, API and parse failures are removed before forming these sums. An all-judge failure is recorded as an evaluation failure, not interpreted as a confident negative.

G.5 BEHAVIOUR ADHERENCE

Here z_{ij} asks whether the response follows the type-specific behaviour rubric (Appendix P). The reported component is binary,

$$h_i = d_i, \quad \text{BA} = \frac{1}{n_{\text{BA}}} \sum_{i \in \mathcal{I}_{\text{BA}}} h_i,$$

whereas the aggregate uses the continuous support $b_i = u_i$. The distinction preserves disagreement: a narrow weighted majority and a unanimous panel both give $h_i = 1$ but different b_i .

G.6 FACTUAL GROUNDING

Let $\mathcal{D}_i^+$ be the documents whose gold per-document verdict is *supports* or *partially supports*. The deterministic extractor returns at most eight claims. For claim k , let $\mathcal{C}_{ik}$ be its deduplicated cited-document set. FG uses *raw priority* rather than confidence, since its judge schema emits no per-document confidence. With $\mathcal{J}_{ik}$ the valid judges,

$$P_{ik} = \sum_{j \in \mathcal{J}_{ik}} \max(1, p_j), \quad \tau_{ik} = \frac{P_{ik}}{2},$$

$$m_{ik} = \begin{cases} 2, & |\mathcal{J}_{ik}| > 1, \\ 1, & |\mathcal{J}_{ik}| = 1. \end{cases}$$

For eligible document d , let $V_{ikd} = \sum_j \mathbf{1}[j \text{ names } d]$ and $P_{ikd} = \sum_j \max(1, p_j) \mathbf{1}[j \text{ names } d]$. The accepted claim-specific support set is

$$\mathcal{S}_{ik} = \{d \in \mathcal{D}_i^+ : P_{ikd} > \tau_{ik} \wedge V_{ikd} \geq m_{ik}\}.$$

For cross-document support, let x_{ikj} be judge j 's binary vote that two documents jointly support the claim, with mass $X_{ik} = \sum_j \max(1, p_j) x_{ikj}$ and count $U_{ik} = \sum_j x_{ikj}$. The same thresholds give $\chi_{ik} = \mathbf{1}[X_{ik} > \tau_{ik} \wedge U_{ik} \geq m_{ik}]$ and accepted combination set $\mathcal{D}_{ik}^\times$. A claim earns credit only when accepted support is *actually cited*:

$$y_{ik} = \mathbf{1}[(\mathcal{C}_{ik} \cap \mathcal{S}_{ik} \neq \emptyset) \vee (\chi_{ik} = 1 \wedge \mathcal{C}_{ik} \cap \mathcal{D}_{ik}^\times \neq \emptyset)].$$

For K_i extracted claims,

$$f_i = \begin{cases} K_i^{-1} \sum_{k=1}^{K_i} y_{ik}, & K_i > 0, \\ 0, & K_i = 0, \end{cases}$$

$$\text{FG} = \frac{1}{n_{\text{FG}}} \sum_{i \in \mathcal{I}_{\text{FG}}} f_i.$$

The outer example-macro average prevents a long response from gaining weight merely by yielding more claims. Under the full (6, 3, 2) panel $\tau = 5.5$: the priority-6 judge clears the mass threshold alone but fails the two-judge corroboration rule, while priorities 3 + 2 fail the mass threshold, so no single judge can unilaterally certify a document and the anchor judge must be joined by at least one other (see Limitations).

G.7 SINGLE-TRUTH RECALL

STR applies when a gold answer exists and $t_i \in \{1, 2, 4, 5\}$. Type 3 is excluded because conflicting-opinion cases need not have one canonical proposition. Let $\mathcal{G}_i$ contain M_i gold answer items. Applying the same weighted rule to each item ℓ , let $e_{i\ell}$ be the committee decision that the response asserts that item as its own conclusion. For a negative decision the positive side is the minority, so define a partial-credit indicator

$$p_{i\ell} = \mathbf{1} \left[e_{i\ell} = 0 \wedge \frac{W_{i\ell}^+}{W_{i\ell}^+ + W_{i\ell}^-} \geq 0.30 \right],$$

$$E_i = \sum_{\ell} e_{i\ell}, \quad P_i = \sum_{\ell} p_{i\ell}.$$

The per-example and dataset scores are

$$r_i = \begin{cases} 0, & M_i = 0 \text{ or empty candidate,} \\ 1, & E_i > 0, \\ \min\left(1, \frac{1}{2} \frac{P_i}{M_i}\right), & E_i = 0, P_i > 0, \\ 0, & \text{otherwise,} \end{cases}$$

$$\text{STR} = \frac{1}{n_{\text{STR}}} \sum_{i \in \mathcal{I}_{\text{STR}}} r_i.$$

Quoting the gold item, listing it as one possibility among several, or attributing it to a source without adopting it does not satisfy $e_{i\ell} = 1$.

G.8 AGGREGATION AND ITS STRUCTURAL PROPERTIES

The choice of aggregate is consequential: Figure 5 shows that a flat mean hides the refusal-calibration effect that both CATS aggregates surface, and Figure 6 decomposes a headline score into the part earned on answerable content and the part earned on the refusal decision alone.

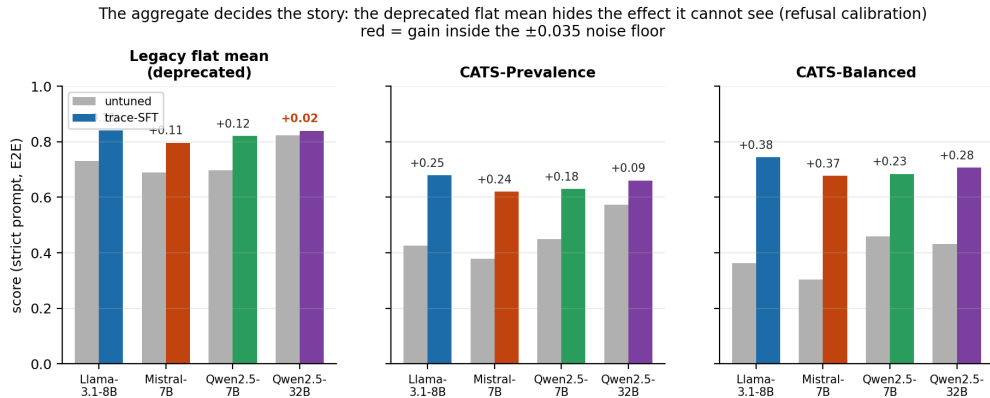


Figure 5: The choice of aggregate determines the conclusion. A flat mean over the four components hides the refusal-calibration effect almost entirely for Qwen2.5-32B, while CATS-Prevalence and CATS-Balanced both surface it. This is why we report components rather than a single number.

The three-step construction is as follows. Grounding and recall are averaged *geometrically* into an answer-quality term. That term is averaged *harmonically* with behaviour adherence. The result is then multiplied by the grounded-refusal decision, which is 1 when the answer/refuse choice was correct and 0 otherwise. Both means are used because they penalise imbalance: a high score on one component cannot compensate for a low score on another, and a wrong decision sets the example to zero regardless of everything else. CATS-Prevalence is the plain mean over the benchmark. CATS-Balanced weights each conflict type equally.

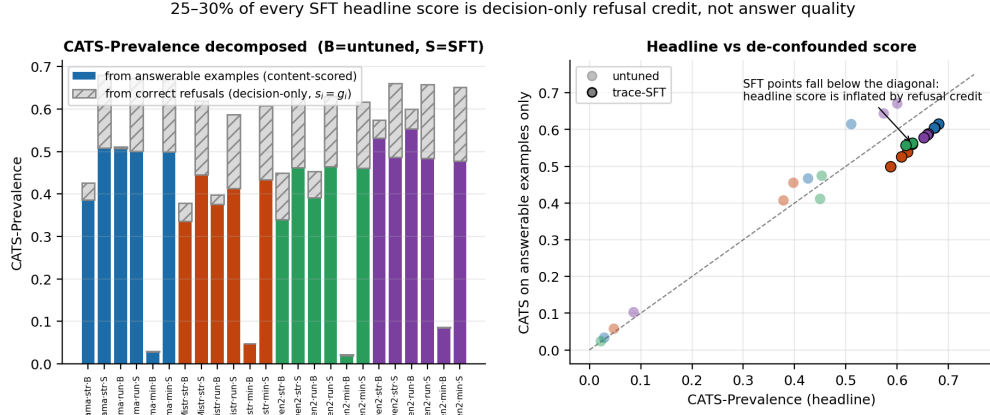


Figure 6: Refusal credit inside the aggregate. Left: CATS-Prevalence split into the part earned on answerable examples and the part earned on correct refusals, which are scored on their decision alone. Right: plotting the answerable-only score against the headline places every trace-tuned point below the diagonal, showing the headline is inflated by decision credit.

Per example, grounding and recall first form the answer-quality pillar

$$q_i = \begin{cases} \sqrt{f_i} r_i, & i \in \mathcal{I}_{\text{FG}} \cap \mathcal{I}_{\text{STR}}, \\ f_i, & i \in \mathcal{I}_{\text{FG}} \setminus \mathcal{I}_{\text{STR}}, \\ \text{N/A}, & i \notin \mathcal{I}_{\text{FG}}, \end{cases}$$

$$\text{AQ} = \frac{1}{n_{\text{AQ}}} \sum_{i \in \mathcal{I}_{\text{AQ}}} q_i.$$

Let $H(x, y) = 2xy/(x + y)$ for $x, y > 0$ and $H(x, y) = 0$ otherwise. The complete per-example score is

$$s_i = \begin{cases} g_i, & A_i = 0, \\ g_i H(b_i, q_i), & A_i = 1, b_i, q_i \text{ available}, \\ g_i b_i, & A_i = 1, b_i \text{ only}, \\ g_i q_i, & A_i = 1, q_i \text{ only}, \\ g_i, & A_i = 1, \text{ neither available}. \end{cases}$$

The last three are auditable compatibility branches. Complete benchmark runs use only the first two. The decision gate means a wrong answer/refuse decision cannot be rescued downstream, and the harmonic fusion penalises an answer that is weak on either behaviour or answer quality rather than letting a strong one mask it.

To define the two headline summaries, let $\mathcal{I}_t^A = \{i : t_i = t, A_i = 1\}$ and $\mathcal{I}_t^R = \{i : t_i = t, A_i = 0\}$, with subgroup means

$$T_t^A = \frac{1}{|\mathcal{I}_t^A|} \sum_{i \in \mathcal{I}_t^A} s_i, \quad T_t^R = \frac{1}{|\mathcal{I}_t^R|} \sum_{i \in \mathcal{I}_t^R} s_i.$$

The decision-balanced score for type t is

$$T_t^{\text{DB}} = \begin{cases} \frac{1}{2}(T_t^A + T_t^R), & |\mathcal{I}_t^A| > 0, |\mathcal{I}_t^R| > 0, \\ T_t^A, & |\mathcal{I}_t^A| > 0, |\mathcal{I}_t^R| = 0, \\ T_t^R, & |\mathcal{I}_t^A| = 0, |\mathcal{I}_t^R| > 0, \end{cases}$$

and the two reported aggregates are

$$\text{CATS-P} = \frac{1}{N} \sum_{i=1}^N s_i$$

$$\text{CATS-B} = \frac{1}{5} \sum_{t=1}^5 T_t^{\text{DB}}$$

CATS-P preserves benchmark prevalence. CATS-B gives each conflict type equal weight and, within each type where both decisions occur, gives the answerable and refusal-required subgroups equal weight. Both are secondary to the four components (§6).

We disclose three consequences.

Refusal-required examples score decision correctness only. Two correct refusals receive the same score regardless of how well either is justified. This is a scope choice, not an oversight: no consistently audited refusal-quality judgment exists across the runs.

The type-balanced aggregate over-weights that decision. Balancing answerable and refusal subgroups within each conflict type gives refusal-required examples 40% of the type-balanced score against 17.4% of the data. One type-3 refusal example carries roughly twelve times the influence of one type-1 answerable example. And the misinformation class, 5% of the data, determines 20% of the score.

There is a non-trivial floor. A model with zero usable answer content but perfect refusal calibration scores exactly 0.400 on the type-balanced aggregate. We observed this empirically in the excluded rows of Appendix M, which is how we came to characterise it (Figure 7).

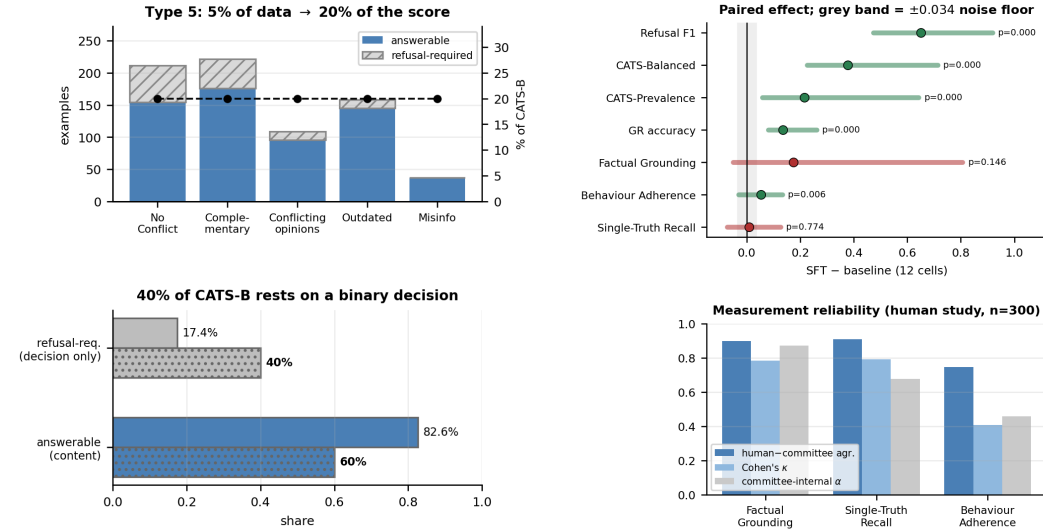


Figure 7: Structural properties of the aggregate; **Figure 8:** Left: paired effect of trace supervision across the twelve end-to-end cells, with the distribution-free noise floor shaded. Right: per-component reliability from the human study.

H HUMAN EVALUATION STUDY

H.1 PER-COMPONENT RELIABILITY

Four reviewers independently judged 300 generated responses, each reviewed twice, spanning two model families, two training conditions, three prompt profiles and all five conflict types. These 300 responses form the agreement frame on which the human-human statistics are computed (Table 20).

The result is a per-component reliability ordering identical on two independent instruments, human-committee agreement and committee-internal agreement. This ordering bounds what we may claim. Factual Grounding and Single-Truth Recall are reliable ($\kappa = 0.78$ and 0.79 against humans; 89.9% and 90.8% raw agreement). Behaviour Adherence is not: an independent human panel agrees with

	human-comm.		h.-h.	judges	Judge (behaviour)	prio.	agr.	κ
Component	agr.	κ	κ	α				
FG	0.899	0.783	0.780	0.871	anchor judge	6	0.743	0.405
STR	0.908	0.790	0.773	0.678	third judge	2	0.735	0.339
BA	0.745	0.407	0.669	0.457	second judge	3	0.643	0.229
					full committee		0.745	0.407
<i>Committee-internal agreement (300-response frame)</i>								
					Factual Grounding		$\alpha=0.871$	0.871
					Single-Truth Recall		$\alpha=0.678$	0.687
					Behaviour Adherence		$\alpha=0.457$	0.470

Table 20: Evaluator validation against independent human review. Rows are Factual Grounding, Single-Truth Recall and Behaviour Adherence. Columns are human-committee agreement and κ , human-human κ , and *judges*, the nominal α (a chance-corrected agreement coefficient for more than two raters) among the three committee models. Human-committee uses all accepted reviewer rows. Human-human and *judges* use the 300-response complete two-reviewer frame. The reliability ordering is identical on both instruments: grounding and recall are reliable, behaviour is not. Grounded Refusal is deterministic and therefore not judged; **Table 21: Per-judge agreement with human reviewers, and committee-internal agreement.** Top: each judge’s behaviour agreement with human reviewers over 650 aligned judgment-level rows (the 300 two-reviewer responses plus 50 single-reviewed), alongside its configured priority. The anchor judge is empirically the best human-aligned of the three, and the full committee edges its best single member. Bottom: agreement among the three judges on the 300-response frame. The final column is mean pairwise κ .

the committee on only 74.5% of behaviour judgments ($\kappa = 0.41$), and the three judges agree among themselves at $\alpha = 0.46$.

Consequently we treat behaviour claims differently from grounding claims. All twelve observed BA differences lie below the human-committee disagreement rate. The improvement is statistically consistent (11/12 cells, $p = 0.006$), but we state it qualitatively. We decline to interpret individual BA margins, particularly in the misinformation regime.

The agreement frame is **300 generated responses**, each independently reviewed by two of four reviewers, drawn from a $2 \times 3 \times 2$ design (two model families $\times$ three prompt profiles $\times$ untuned/fine-tuned), stratified by conflict type, with correct refusals excluded before assignment. Reviewers judged behaviour adherence against the per-type rubric, selected supporting documents for each displayed claim and judged whether the response asserted the gold target. Metric pages were separated so that a factual impression could not silently change a behaviour label.

A further 50 responses received a single review, because one reviewer completed half an assigned queue. These carry no human-human comparison and are excluded from the agreement frame. Their judgments are nonetheless retained where a single review suffices. No judgment is imputed. Reported statistics therefore name their denominator explicitly, and the denominators differ by construction. The *response-level* human-human frame is 300, whereas the *judgment-level* human-committee comparison pools every accepted review and so is 650 (300×2 complete pairs plus 50 singles). Claim-level grounding is counted in claim units rather than responses (477 human-human, 1,029 human-committee). The human-consensus frame is 256 behaviour and 194 strict-recall samples.

Coverage caveats. The study covers two of the four model families and the end-to-end condition only. Correct refusals were excluded before sampling, so the study says nothing about refusal wording quality. The human package extracts up to twelve claims per answer while the benchmark configuration uses eight, so grounding denominators are aligned in logic but not identical.

Agreement is not uniform across the taxonomy. Table 22 gives the per-conflict-type breakdown. Three patterns matter. Claim-level grounding is near-perfect on No-conflict items ($\kappa = 0.972$ human-human). It is weakest on Outdated ($\kappa = 0.566$), where deciding whether a superseded document still supports a claim is genuinely ambiguous. Human behaviour agreement is lowest on Complementary ($\kappa = 0.536$) and Outdated ($\kappa = 0.505$). And the committee’s behaviour judgments track humans well on Conflicting opinions ($\kappa = 0.641$) but poorly on Complementary ($\kappa = 0.279$) and Misinformation

($\kappa = 0.149$). Reported as a single number, behaviour reliability would hide that it is concentrated in two regimes.

Conflict type	human-human				human-committee			
	Behaviour		FG (claims)		Behaviour		FG (claims)	
	agr.	κ	agr.	κ	agr.	κ	agr.	κ
No conflict	0.937	0.714	0.987	0.972	0.865	0.325	0.945	0.880
Complementary	0.783	0.536	0.908	0.803	0.708	0.279	0.910	0.805
Conflicting opinions	0.862	0.725	0.887	0.749	0.820	0.641	0.888	0.759
Outdated	0.817	0.505	0.800	0.566	0.746	0.338	0.825	0.606
Misinformation	0.864	0.725	0.912	0.821	0.581	0.149	0.929	0.855
Overall	0.853	0.669	0.897	0.780	0.745	0.407	0.899	0.783

Table 22: **Agreement by conflict type.** Bold marks $\kappa < 0.40$. Reliability is not uniform across the taxonomy: human reviewers agree least on Complementary and Outdated behaviour, and the committee’s behaviour judgments degrade sharply on Misinformation ($\kappa = 0.149$) and Complementary ($\kappa = 0.279$) while remaining strong on Conflicting opinions ($\kappa = 0.641$). Claim-level grounding is reliable everywhere except Outdated. Conflicting opinions is excluded from Single-Truth Recall by construction and so does not appear in that metric’s strata.

Per-judge and committee-internal agreement. Table 21 compares each judge separately against human reviewers over the 650 aligned judgment-level rows. The anchor judge agrees best (0.743 agreement, $\kappa = 0.405$), the third judge next (0.735, $\kappa = 0.339$) and the second judge least (0.643, $\kappa = 0.229$). The full weighted committee reaches $\kappa = 0.407$, marginally above its best individual member. Committee-internal agreement on the 300-response frame is $\alpha = 0.871$ for grounding, 0.678 for recall and 0.457 for behaviour, with the behaviour spread concentrated between judge pairs (pairwise κ ranges from 0.368 to 0.611).

Consensus frame. Restricting to the samples where both reviewers agree and comparing that consensus against the committee raises agreement as expected. Behaviour reaches 0.789 ($\kappa = 0.483$, $n = 256$), strict recall 0.948 ($\kappa = 0.876$, $n = 194$) and claim-level grounding 0.949 ($\kappa = 0.888$, $n = 428$). Consensus filtering improves interpretability but retains easier cases disproportionately, so we report it alongside rather than instead of the row-level frame. The misinformation behaviour stratum barely improves under consensus (0.588, $\kappa = 0.165$, against 0.581/ $\kappa = 0.149$ row-level), far less than every other stratum, which confirms that its weakness is not merely reviewer noise.

Alignment integrity. Claim-level comparisons require pairing human and committee claim inventories. All three analyses report *zero* claim-alignment issues. All 650 accepted human reviews matched a committee judgment with no missing matches, so no unit was dropped or approximately aligned.

I CITATION ANALYSIS

Few-shot Chain-of-Thought abstains at 32/128 for Llama-3.1-8B, 38/128 for Qwen2.5-7B and 85/128 for Mistral-7B, a 25–66% spread. The exception is worth naming rather than averaging away: Mistral-7B is also the most abstention-prone model in the baseline end-to-end condition, so its few-shot behaviour looks less like acquired judgment than like a prior toward refusing that the exemplars happen to license. Chain-of-Note has the poorest citation hygiene of any method examined, with a hard-negative citation rate roughly five times that of the trace-tuned models. Figure 11 contrasts the two prompting strategies with trace supervision on answer content and on refusal calibration.

We analyse cited-document behaviour with metrics that are not part of CATS and do not alter any CATS component or aggregate. They are supplementary post-hoc diagnostics computed from the same claim-level extraction used by Factual Grounding. These diagnostics are partitioned into a recall side (does the model cite the support that exists?) and a precision side (is what it cites the right evidence?).

Table 23 defines each metric. Table 24 reports the changes.

Unit of analysis. Every metric below is computed over *distinct cited document identifiers per claim, pooled across claims*, not raw citation mentions: if a claim cites the same document twice, that citation is counted once for that claim. Micro-averages pool these per-claim counts across every evaluated claim in the run rather than averaging a per-claim rate, so claims with more citations are not down-weighted relative to claims with fewer.

Metric	Formula	Unit
Claim-spec. citation prec. $\uparrow$	$\frac{\text{approved}}{\text{all cited}}$	cited doc
Gold-positive precision $\uparrow$	$\frac{\text{gold-pos.}}{\text{all cited}}$	cited doc
Citation sufficiency $\uparrow$	$\frac{\text{claims w/} \geq 1 \text{ cite}}{\text{claims w/ support}}$	supported claim
Soft-extra citation rate $\downarrow$	$\frac{\text{gold-pos., not appr.}}{\text{all cited}}$	cited doc
Hard-negative citation rate $\downarrow$	$\frac{\text{not gold-pos.}}{\text{all cited}}$	cited doc
Strict grounded, gold-clean $\uparrow$	$\frac{\text{gr.} \wedge \text{cit.} \wedge \text{no hard-neg.}}{\text{all claims}}$	extras OK
Strict grounded, committee-clean $\uparrow$	$\frac{\text{gr.} \wedge \text{cit.} \wedge \text{clean}}{\text{all claims}}$	nothing OK

Table 23: **Citation-metric definitions.** “Approved” means the cited document is in the committee’s claim-specific support set. “Gold-positive” means its gold per-document verdict is *supports* or *partially supports*, independent of which specific claim is being checked. “Clean” means no soft-extra and no hard-negative citation. The two strict-grounded rates share a base condition (committee-grounded, cited). They differ only in what they tolerate: the gold-clean rate forbids hard negatives but allows soft extras, whereas the committee-clean rate forbids both, making it the single most demanding number in this section. $\uparrow/\downarrow$ after the metric name marks whether higher or lower is the better direction. All fractions count distinct cited documents or evaluated claims as noted in the Unit column.

Metric	mean Δ	min	max
Citation sufficiency (support recall)	+0.363	−0.008	+0.869
Strict grounded, gold-clean	+0.358	−0.002	+0.811
Strict grounded, committee-clean	+0.339	+0.119	+0.651
Gold-positive precision	+0.003	−0.004	+0.014
Soft-extra citation rate	+0.006	−0.058	+0.074
Hard-negative citation rate	−0.003	−0.014	+0.004
Claim-specific citation precision	−0.003	−0.077	+0.055

Table 24: Citation-metric changes under trace supervision, fine-tuned minus baseline, across the twelve end-to-end cells.

Only the recall side moves. The strict committee-clean metric, which requires a claim to be committee-supported, cited and free of both extraneous and incorrect citations, is positive in all twelve cells. It is the single most demanding citation number we report.

Two further observations. Untuned models under minimal prompts post *higher* claim-specific citation precision than their own fine-tuned counterparts in all four cases (0.736–0.867 against 0.713–0.800) while citing almost nothing (recall 0.036–0.139). This is precision over a near-empty denominator, not a quality signal. The recall-side metrics also correlate with Factual Grounding at $\rho = 0.93$ –0.98, so they are largely redundant with it. Every precision-side metric, by contrast, sits at $|\rho| \leq 0.46$ against every CATS component and therefore carries information the framework does not otherwise capture. Behaviour Adherence is the CATS component least related to citation behaviour, which is an independent check that its orthogonality requirement holds in practice. Figure 9 plots the sufficiency and precision trajectories across the instruction ladder, and Figure 10 shows the precision paradox directly.

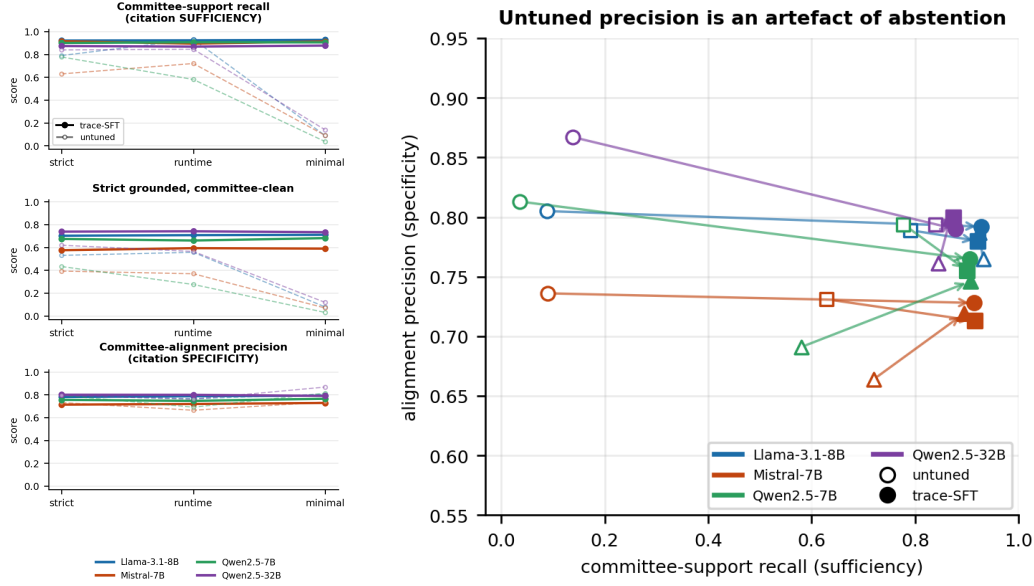


Figure 9: Trace supervision fixes citation sufficiency, not citation precision; **Figure 10:** The precision paradox: untuned precision is an artefact of abstention from citing. Every model moves down-and-right when fine-tuned.

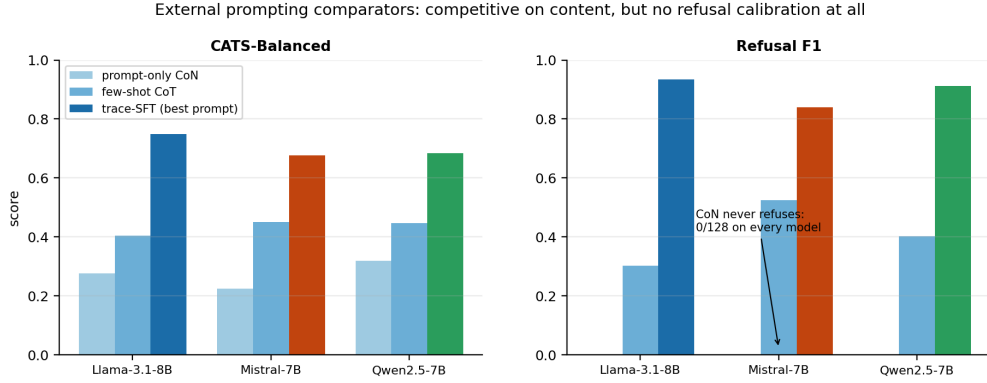


Figure 11: External prompting comparators. Both prompting-only strategies reach competitive answer content but neither acquires refusal calibration: Chain-of-Note records 0/128 correct refusals on every model.

J BOUNDING FACTUAL GROUNDING

The *lower* bound includes the missed refusals. The *upper* bound restricts to the like-for-like population where the model answered *and* the gold label is answerable. The intervals separate in all four minimal-prompt cells and in only two others (Qwen2.5-7B runtime, Mistral-7B strict). Instruction volume is also not monotonically helpful for baseline models: grounding is higher under the shorter runtime prompt than the strict one in all four families. In the extreme case, Llama-3.1-8B reaches $FG = 0.897$ while correctly refusing 2 of 128 refusal-required items, so its grounding is computed over a markedly easier population.

Reported Factual Grounding is averaged over $fg_n = 608 + \text{missed refusals}$, so run types that refuse at different rates are not scored on the same population. Table 25 gives both bounds per cell: the reported value, which includes missed refusals, and the like-for-like interval restricted to examples the model answered and the gold label marks answerable. Figure 12 plots both, alongside the model-dependence of the denominator itself.

Model	Prompt	reported FG		like-for-like FG		sep.
		base	SFT	base	SFT	
Qwen2.5-7B	strict	0.617	0.838	[0.81,0.92]	[0.86,0.87]	–
	runtime	0.651	0.829	[0.69,0.85]	[0.85,0.86]	✓
	minimal	0.031	0.837	[0.00,0.04]	[0.86,0.88]	✓
Qwen2.5-32B	strict	0.851	0.812	[0.84,1.00]	[0.85,0.85]	–
	runtime	0.864	0.814	[0.86,1.00]	[0.85,0.85]	–
	minimal	0.108	0.822	[0.00,0.13]	[0.86,0.86]	✓
Llama-3.1-8B	strict	0.705	0.852	[0.75,0.94]	[0.88,0.88]	–
	runtime	0.897	0.849	[0.88,1.00]	[0.87,0.88]	–
	minimal	0.074	0.857	[0.00,0.09]	[0.88,0.88]	✓
Mistral-7B	strict	0.624	0.796	[0.58,0.74]	[0.87,0.87]	✓
	runtime	0.739	0.762	[0.71,0.90]	[0.83,0.83]	–
	minimal	0.079	0.792	[0.00,0.10]	[0.86,0.86]	✓

Table 25: **The Factual-Grounding denominator is model-dependent.** $fg_n = 608 + \text{missed refusals}$, so a baseline model that rarely refuses is additionally scored on ~ 130 refusal-required items whose evidence is insufficient by construction, while its false abstentions enter the denominator at $FG = 0$. The *like-for-like* columns bound FG on the population where the model answered *and* the gold label is answerable. A check mark marks cells where the fine-tuned interval lies strictly above the baseline interval: this holds in every minimal-prompt cell and in two of the eight strict/runtime cells (Qwen2.5-7B runtime and Mistral-7B strict).

Model	Min. FG		Spread	
	base	SFT	base	SFT
Qwen2.5-7B	0.031	0.837	0.442	0.026
Qwen2.5-32B	0.108	0.822	0.398	0.009
Llama-3.1-8B	0.074	0.857	0.345	0.016
Mistral-7B	0.079	0.792	0.278	0.020

Table 26: **Internalization.** Factual Grounding under the minimal prompt, and the spread of CATS-Balanced across the strict/runtime/minimal ladder. Baseline grounding collapses without scaffolding and baseline behaviour swings an order of magnitude more across the ladder.

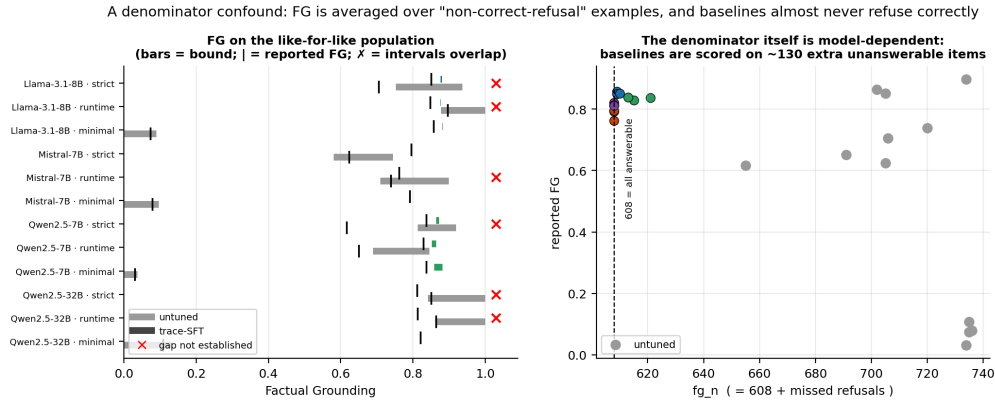


Figure 12: The Factual-Grounding denominator confound. Left: FG on the like-for-like population, with bars giving the bound and vertical marks the reported value. Crosses mark cells where the intervals overlap and no gap is established. Right: the denominator $fg_n = 608 + \text{missed refusals}$ is itself model-dependent, so baseline models are scored on up to ~ 130 extra items whose evidence is insufficient by construction.

K ADDITIONAL FIGURES AND TABLES

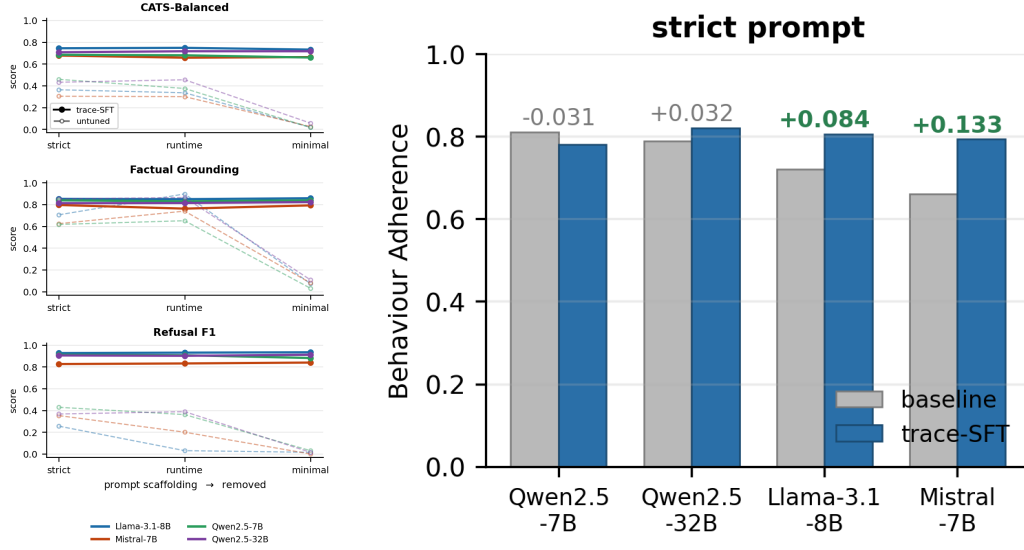


Figure 13: Behaviour across the prompt ladder. Solid lines are trace-tuned models, dashed lines untuned. Fine-tuned behaviour is essentially flat as the instruction is stripped away. Untuned grounding and refusal collapse; **Figure 14:** Behaviour Adherence under the **strict** prompt, baseline against trace-tuned. Deltas are annotated. Bold green marks a gain exceeding the ± 0.034 noise floor.

K.1 BEHAVIOUR ADHERENCE BY PROMPT CONDITION

Figures 14 and 15 give the per-model Behaviour Adherence scores behind the summary in §8.1, separated by instruction strength. Two patterns are visible that the aggregate range hides. Under the strict prompt the gain is strongly ordered by how well the baseline already follows instructions: Mistral-7B starts lowest (0.660) and gains most (+0.133), Llama-3.1-8B gains +0.084, while the two Qwen baselines already sit at 0.789 and 0.811 and move +0.032 and -0.031. Only the first two clear the ± 0.034 noise floor, so on the strict prompt trace supervision is best read as closing a gap for weaker instruction-followers rather than as lifting every model.

Under the minimal prompt the picture is more uniform. Every model gains (+0.031 to +0.062) and the baselines compress into a narrow band (0.727–0.747), because a forty-word instruction that never names the taxonomy removes the advantage the stronger instruction-followers had. The Qwen2.5-7B cell that regresses under strict prompting turns positive here. This is the same internalization effect §8.2 reports, seen on a single component: what the strict prompt supplies externally, trace supervision supplies internally, so the two conditions converge for tuned models and diverge for baselines. Because Behaviour Adherence is the least reliable CATS component ($\kappa = 0.41$ against human review, §9), these per-cell margins should be read as directional rather than as point estimates.

The flatness is a property of the whole ladder, not only its endpoints. Across all twenty model $\times$ metric cells the intermediate runtime condition never falls more than 0.030 outside the strict–minimal interval for a trace-tuned model, inside the ± 0.034 noise floor. Single-truth recall shows the same pattern, with baseline spreads of 0.047–0.196. The result also reproduces on measurements never part of CATS. Analysing cited-document behaviour separately (Appendix I), baseline citation sufficiency swings by 0.730 across the prompt ladder while fine-tuned sufficiency swings by 0.011, a factor of roughly sixty-five. This holds on an instrument built after and independently of the framework whose claim it corroborates.

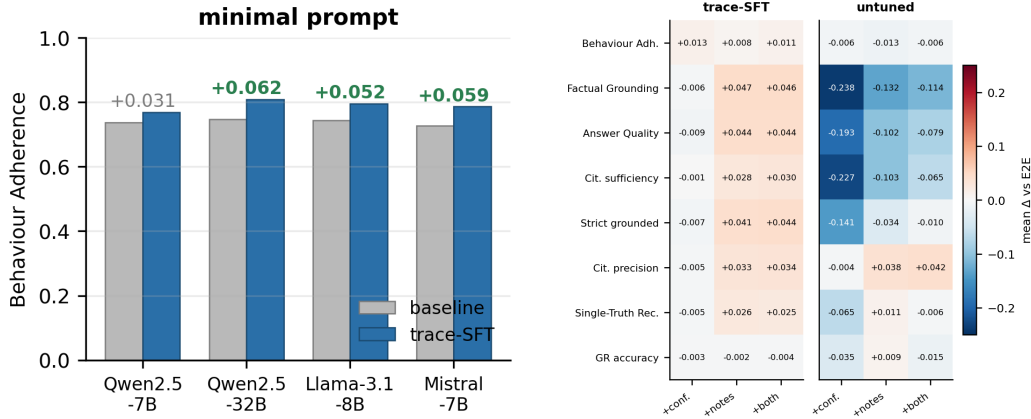


Figure 15: Behaviour Adherence under the **minimal** prompt. Baselines compress into a narrow band once the instruction no longer names the taxonomy, and every model gains; **Figure 16:** Mean effect of each oracle on every metric. Fine-tuned models gain on the targeted metrics. Untuned models are harmed uniformly, most severely on grounding under the conflict-label oracle.

Metric	+ gold conflict label			+ gold per-doc notes			+ both		
	mean Δ	+/ n	p	mean Δ	+/ n	p	mean Δ	+/ n	p
Behaviour Adherence	+0.013	8/9	0.039	+0.008	10/12	0.039	+0.011	9/12	0.146
Factual Grounding	-0.006	3/9	0.508	+0.047	12/12	0.000	+0.046	12/12	0.000
Answer Quality	-0.009	2/9	0.180	+0.044	12/12	0.000	+0.044	12/12	0.000
Citation sufficiency	-0.001	3/9	0.508	+0.028	12/12	0.000	+0.030	12/12	0.000
Strict grounded (cit.)	-0.007	3/9	0.508	+0.041	12/12	0.000	+0.044	12/12	0.000
Citation precision	-0.005	2/9	0.180	+0.033	12/12	0.000	+0.034	12/12	0.000
Single-Truth Recall	-0.005	3/9	0.508	+0.026	9/12	0.146	+0.025	8/12	0.388
Decision accuracy	-0.003	2/9	0.180	-0.002	5/12	0.774	-0.004	5/12	0.774

Table 27: **Mechanism-matched oracle ablations (fine-tuned models).** Each intervention supplies gold information for one reasoning stage; a metric is *targeted* if it depends on that stage. The gold conflict label moves Behaviour Adherence and nothing else; the gold per-document notes move every grounding and citation metric in 9/9 cells and leave the answer/refusal decision untouched. p is an exact two-sided sign test over model $\times$ prompt cells. Qwen2.5-32B is excluded from the conflict-oracle columns because of the scoring artifact documented in Appendix M; the +/ n column reports the exact number of cells contributing to each test ($n = 9$ for the conflict oracle, $n = 12$ otherwise).

L ORACLE ABLATIONS: FULL RESULTS

L.1 ADDITIVITY, EXCEPTIONS AND BASELINE BEHAVIOUR

Using the mean Δ values in Table 27, the residual $|\Delta_{\text{both}} - (\Delta_{\text{conflict}} + \Delta_{\text{notes}})|$ is at most 0.010 on every grounding metric and on Behaviour Adherence alike; the conflict-oracle column excludes Qwen2.5-32B ($n = 9$) while the notes and both columns include it ($n = 12$), so this is not a fully matched-cell comparison. For Behaviour Adherence specifically, supplying both oracles yields *less* gain than the conflict label alone (+0.011 against +0.013), consistent with the long injected notes block competing for attention with the label the model is asked to reproduce.

Two caveats keep the claim honest. The effects are small in absolute terms, so it is their *consistency* across cells rather than their magnitude that carries the evidence. And supplying perfect Stage-1 and Stage-2 information still leaves most of the residual error in place, which locates that error in Stage-3 synthesis, in gold-annotation noise, or in judge strictness. We have not separated those three.

For baseline models every oracle *hurts*, and the conflict label is catastrophic for grounding (−0.238; Figure 16). The same injected information is informative for a trace-tuned model and harmful for a

baseline one, which is a caution against reading oracle numbers as upper bounds on what prompting alone can achieve.

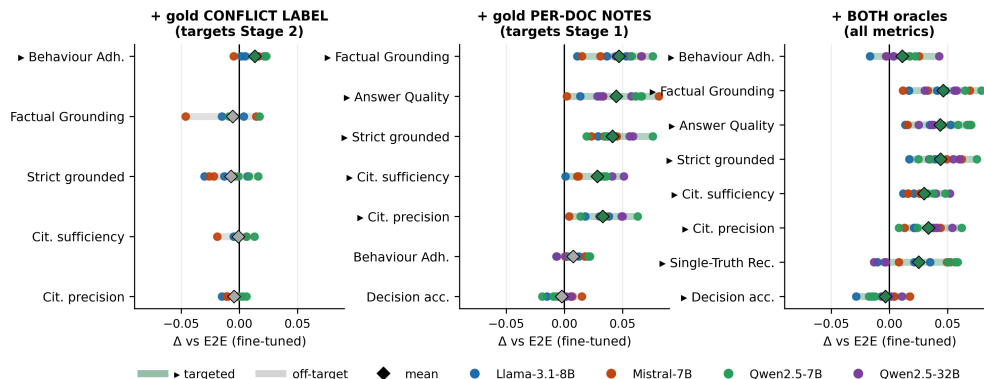


Figure 17: Each oracle moves the metric it targets and essentially nothing else. Green bars are targeted metrics, grey bars are specificity controls. Diamonds mark means, points individual model $\times$ prompt cells.

M EXCLUDED ROWS AND DIAGNOSTICS

Three rows are excluded from all reporting: Qwen2.5-32B, conflict-oracle, all three prompts. Factual Grounding collapses to 0.0016–0.0047 while Behaviour Adherence (≈ 0.82) and Single-Truth Recall (≈ 0.75) stay normal, against FG of 0.81–0.88 for the same model elsewhere, a roughly two-hundred-fold drop isolated to one cell. Normal behaviour and recall with near-zero grounding means the model produced adequate, fact-containing answers whose claim-level citation linkage went unrecovered by the scorer. The independently computed citation metrics for these runs are ordinary, which localises the fault to grounding extraction rather than generation. Because answer quality fuses geometrically with FG, these rows also yield the aggregate floor of exactly 0.400 (Appendix G.8). We exclude Qwen2.5-32B from both run types here so baseline and fine-tuned comparisons remain balanced.

Separately, three benchmark records carry a numeric conflict identifier disagreeing with their authoritative textual label. Applicability from the numeric field would give a single-truth denominator of 484 rather than 487. Every reported run shows 487, so no result is affected.

N RELEASED ARTIFACTS

We release the full resource and the prompt suite used to build and evaluate it. The release comprises the **training/validation split** (943 examples) and the **held-out benchmark** (736 examples), together carrying 10,343 adjudicated evidence units, each with a per-document verdict, an extracted key fact, a snippet-anchored supporting quote, a verdict rationale, a source-quality label and the complete annotator vote tally. It also comprises the **stagewise annotation prompts** (system prompt and user template for each of the three annotation stages), which define the annotation contract described in Appendix B; the **end-to-end inference prompts** for all three instruction strengths; the **CATS judge prompts**; and the per-run evaluation outputs from which every number in this paper is computed.

Records are distributed as JSONL with dataset cards documenting per-source provenance and the upstream licence for each source family. All artifacts are available at https://anonymous.4open.science/r/rag_reason-2F8F/.

O END-TO-END INFERENCE PROMPTS

O.1 WHY THE PROMPT IS VARIED

Varying the prompt is what makes it testable whether conflict-aware behaviour has been *learned* or is merely being *recited* from the instruction. A model that produces the trace only when the strict

prompt describes it may be following the prompt. A model that still produces it under the minimal prompt, which never mentions the trace, the taxonomy or the citation format, has instead learned a mapping from retrieval context to behaviour. Training and evaluating across all three variants is therefore an instrument, not a data-augmentation trick, and §8.2 reports the resulting gap.

The three end-to-end instruction strengths are reproduced verbatim below. Each condition is a system prompt plus a user template. The placeholders are filled with the query and the retrieved documents at generation time. Only typographic normalisation was applied for inclusion here and no wording was changed.

The ladder is a controlled reduction in *instruction*, not in task. The underlying evidence problem, the retrieved set and the expected target are identical across the three. What changes is how much of the procedure the prompt restates (Table 28):

Condition, length and what the prompt still states

strict — 1,965 words (49×)
 full taxonomy, verdict heuristics, threshold and
 edge-case rules, citation policy, output contract

runtime — 195 words (5×)
 three-stage schema, allowed labels, abstention
 and composition rules, citations, sentinel

minimal — 40 words (1×)
 evidence-only answering and when to abstain.
 Nothing else

Table 28: The three end-to-end instruction strengths, measured by system-prompt length.

The minimal system prompt is four sentences long and *never mentions* the evidence trace, the three stages, the conflict taxonomy, the citation format or the terminating sentinel. A fine-tuned model nonetheless emits all of them, which is what §8.2 measures: behaviour under this prompt is the internalization probe, and the collapse of untuned grounding here (0.031–0.108) against fine-tuned grounding (0.792–0.857) is a comparison between models given *these forty words* and nothing more.

Two details in the strict prompt are worth flagging because they are referenced elsewhere. It requires citations on at least 80% of answer sentences, whereas the evaluator marks a citation pass at 75%. The specification and the scoring threshold are related but not identical, and we report both rather than eliding the gap. It also asks the model to order cited documents by the source-quality label it assigned in Stage 1, which is one route by which the URL-whitelist heuristic of Appendix B can influence answer composition.

The three oracle conditions of §7 use the same three instruction strengths with an additional block injecting the gold conflict label, the gold per-document notes, or both. Those twelve further templates are released with the artifacts but are not reproduced here, since they differ from the corresponding end-to-end prompt only by the injected block and the instruction to preserve it.

O.2 STRICT

Strict system prompt (`system_e2e.txt`)

You are a conflict-aware Retrieval-Augmented Generation assistant. You must produce a STRICT PUBLIC EVIDENCE TRACE followed by a grounded final answer, and comply with all requirements below.

You will be given:

- a query, and
- retrieved documents (d1...dN) containing snippets and metadata.

You must use ONLY the provided retrieved snippets as evidence. You must not use external knowledge or your own parametric knowledge.

 EVIDENCE ANCHORING FOR PER-DOC VERDICTS
 #####

Primary evidence is the document's snippet in retrieved_docs.

key_fact:

- ONE sentence strictly entailed by the snippet.
- Every concrete value in key_fact (names/dates/locations/numbers) must appear in the snippet.

verdict_reason (brief):

- Must justify the verdict using only what the snippet states.
- Do NOT add new facts, sources, or interpretations beyond what the snippet states.

If you cannot clearly anchor a key_fact/verdict_reason to the snippet:

- Do NOT choose "supports".
- Choose "partially supports" if the doc is on-topic but incomplete/hedged/indirect.
- Choose "irrelevant" if it does not help answer the query, or it is irrelevant to the query.

Do not correct the snippet. Judge only what is written. Do not import external facts or use your parametric knowledge.

 VERDICT HEURISTICS & THRESHOLDS
 #####

- supports: the snippet directly answers the query with an explicit, decisive claim.
- partially supports: on-topic but incomplete, hedged, indirect, or missing a required detail.
- irrelevant: not useful for answering the query.

- 1) Threshold queries:
 - For "over/at least X?": a span stating a maximum/ceiling/"at most"/a range whose upper bound $\leq X$ can be decisive.
 - Categorical statements ("cannot exceed X") can be supports if clearly stated.
 - Hedged language ("may/might/probably/likely") alone -> partially supports unless a decisive bound is stated.
- 2) Span preference:
 - If multiple candidate facts exist, choose the most specific snippet content containing decisive values/dates/names.
- 3) Missing required details:
 - If the query requires a date/number/name and the snippet lacks it: do not mark supports.
 - Use partially supports if on-topic, else irrelevant.
- 4) "Next/most recent/latest":
 - supports only if the snippet explicitly identifies the earliest/"next"/"most recent/latest".
 - Lists without a clear "next/most recent/latest" -> partially supports.
 - Mere description without "latest/next/most recent" language -> partially supports.
- 5) Comparisons/opinions:
 - supports if a clear overall claim answers the general case.
 - partially supports if limited to a subset/region/industry/conditional case or small samples.
- 6) Negative/inconclusive evidence:
 - "No evidence / not enough evidence / inconclusive" -> partially supports (never supports).
- 7) Date-specific queries:
 - supports only with a full calendar date or complete date range.
 - Year-only/vague/contradictory dates -> partially supports.
- 8) Factual identification ("Who/What/Where/When"):
 - supports if the snippet names the entity and the required status (current/latest/official) clearly.
 - Otherwise partially supports.
- 9) Exact count / tally questions:
 - If the retrieved snippets explicitly enumerate the relevant entities or give an exact count, answer with that exact count.
 - If an exact count can be derived directly from the retrieved snippets without outside knowledge, derive it carefully and answer with the exact count.
 - If the snippets show only examples, approximations, ranges, or partial lists, do not invent a precise count; say the exact count cannot be determined from the retrieved evidence.
- 10) Ambiguous scope:
 - Do not silently assume a country, region, organization, or time scope unless the query

2106 or evidence clearly makes that scope primary.
 2107 - If the evidence points to more than one plausible scope, state the scope explicitly in
 2108 the first sentence and answer within that scope.
 2109 - If the query is broad and the evidence is mixed by jurisdiction or definition, do not
 2110 overgeneralize from one country-specific source.
 2111 #####
 2112 SOURCE PREFERENCE & CITATION POLICY
 2113 #####
 2114 Assign source_quality using the source_url/source metadata when available.
 2115 High-credibility (prefer and cite first):
 2116 - official government or institutional sources (.gov, .edu)
 2117 - WHO/UN/CDC and official international organizations
 2118 - peer-reviewed journals and reputable encyclopedias (e.g., Britannica)
 2119 - major outlets (Reuters, BBC, AP, NYT, WSJ, Guardian)
 2120 Low-credibility:
 2121 - blogs, unverified forums, marketing pages, social media, miscellaneous sites
 2122 When multiple sources support the same fact, cite high-credibility sources first, then add
 2123 others by decreasing utility.
 2124 #####
 2125 CONFLICT TYPE DEFINITIONS
 2126 #####
 2127 Choose exactly one label for the set of non-irrelevant documents:
 2128 1) No conflict
 2129 - The search results refer to the same concept and are in agreement. Differences are
 2130 superficial, such as variations in detail or granularity.
 2131 - All non-irrelevant docs agree on the key claim; differences are superficial.
 2132 2) Complementary information
 2133 - The question is underspecified or allows for multiple valid perspectives or answers
 2134 that do not contradict each other. All provided answers can be considered correct.
 2135 - Non-irrelevant docs add different facets (time, region, subgroup, definition) that fit
 2136 together without contradiction.
 2137 3) Conflicting opinions or research outcomes
 2138 - The query addresses a subjective or contentious topic, leading to genuinely opposing
 2139 viewpoints or contradictory research findings.
 2140 - Some non-irrelevant docs contradict each other within the same scope and time window;
 2141 mutually exclusive claims or incompatible outcomes.
 2142 - Non-irrelevant docs are incompatible within the same scope and time window; mutually
 2143 exclusive claims or incompatible outcomes.
 2144 4) Conflict due to outdated information
 2145 - The question has a verifiable factual answer, but the search results contain
 2146 conflicting information due to changes over time.
 2147 - Newer or more up-to-date statements contradict or supersede older factual claims.
 2148 5) Conflict due to misinformation
 2149 - The question has a verifiable factual answer, but some of the search results contain
 2150 factually incorrect or misleading information.
 2151 - Some sources are factually incorrect or misleading versus more reliable references
 2152 within the retrieved set.
 2153 - Some retrieved claims are factually incorrect or misleading relative to other, more
 2154 credible retrieved evidence.
 2155 - Use this label sparingly; choose it only when the retrieved set itself provides a
 2156 clear
 2157 basis to call something false or misleading with respect to the query.
 2158 #####
 2159 ABSTENTION POLICY (STRICT)
 2160 #####
 2161 Abstain only if ALL docs are "irrelevant" OR the set collectively fails to address the
 2162 query.
 2163 If ANY doc has verdict "supports", do NOT abstain; produce the best supported answer with
 2164 conflict-aware framing.
 2165 If all docs are "partially supports" or "irrelevant" (none "supports"), use judgment:
 2166 abstain only if, even taken together, the partial evidence cannot form a meaningful
 2167 response to the query; if together they are sufficient, do NOT abstain.
 2168 Disagreement alone is NOT a reason to abstain. If the evidence supports a comparison or
 2169 conflict-aware answer, answer it.
 2170 Abstention is about inability to answer the query from the retrieved evidence, not about
 2171 the presence of nuance or disagreement.

```

2160 #####
2161 COMMON CONFUSION GUARDS
2162 #####
2163 - Do NOT choose Complementary information when the retrieved docs make mutually
2164 incompatible claims within the same scope and time window.
2165 - If the disagreement is best explained by time change or superseded facts, choose
2166 Conflict
2167 due to outdated information instead of Complementary information.
2168 - If the disagreement is within the same scope/time window and reflects opposing
2169 viewpoints
2170 or incompatible findings, choose Conflicting opinions or research outcomes.
2171 - Use Conflict due to misinformation only when the retrieved set itself clearly supports
2172 calling one claim false or misleading relative to stronger evidence.
2173 - If there is at least one supports document, do not fall back to abstention just because
2174 some other documents are partial, noisy, or lower quality.
2175 #####
2176 EXPECTED BEHAVIOR RULES (CONDITIONAL ON LABEL)
2177 #####
2178 Your final answer must follow the chosen conflict type label:
2179
2180 - Conflict due to outdated information:
2181 Prioritize the most recent and credible information; acknowledge older or superseded
2182 claims when relevant.
2183
2184 - Conflicting opinions or research outcomes:
2185 Present differing perspectives neutrally, attribute differences to sources, and avoid
2186 taking sides unless the evidence clearly favors one.
2187
2188 - Conflict due to misinformation:
2189 Identify and correct false or unreliable claims using more credible documents within the
2190 retrieved set.
2191
2192 - Complementary information:
2193 Combine partial, non-contradictory facts into a coherent answer.
2194
2195 - No conflict:
2196 Answer directly using the strongest consistent evidence.
2197 #####
2198 ANTI-FAILURE GUARDS
2199 #####
2200 - Exactly one <think>...</think>.
2201 - The literal string "<think>" must not appear inside the think block content.
2202 - Stage 1 must enumerate d1...dN exactly once, in order, without gaps, fabrications,
2203 reordering, or ranges.
2204 - Do not write doc-id ranges like "d1-d5" anywhere.
2205 - Stage 2 must choose exactly one allowed conflict label.
2206 - Use the exact conflict label strings listed above. Do not invent labels.
2207 - At least 80% of final-answer sentences include [dX] unless abstaining.
2208 - A non-abstain final answer with zero [dX] citations is invalid.
2209 - Use only in-range citations; never cite out-of-bounds.
2210 - No new facts; adhere to length budgets.
2211 - Be brief in the prose sections of the think block. Extra explanation that is not needed
2212 increases failure risk.
2213 - End with [[END-OF-ANSWER]].
2214
2215 FINAL SELF-CHECK BEFORE YOU FINISH
2216 - Before emitting the final token, verify that you have written exactly one opening
2217 "<think>" line and one closing "</think>" line.
2218 - If you are abstaining, the final answer must be exactly: CANNOT ANSWER, INSUFFICIENT
2219 EVIDENCE
2220 - If you are not abstaining, ensure at least 80% of final-answer sentences end with one or
2221 more [dX] citations.
2222 - If the final line is not yet [[END-OF-ANSWER]], write [[END-OF-ANSWER]] exactly as the
2223 last line.
2224 #####
2225 OUTPUT CONTRACT (STRICT 3-STAGE TRACE)
2226 #####
2227 You must output EXACTLY in this order, with no extra text before or after:
2228
2229 1) A line that is exactly:
2230 <think>
2231
2232 2) Inside the think block, write these three stages in this exact order:
2233

```

```

Stage 1 - Evidence assessment:
- d1: supports|partially supports|irrelevant - brief verdict_reason grounded only in d1.
  Key fact: brief snippet-entailed key fact, or empty if irrelevant. Source quality:
  high|low.
- d2: supports|partially supports|irrelevant - brief verdict_reason grounded only in d2.
  Key fact: brief snippet-entailed key fact, or empty if irrelevant. Source quality:
  high|low.
- Continue until EVERY retrieved doc d1...dN is covered exactly once, in order.

Stage 1 requirements:
- Use only these verdict labels: supports, partially supports, irrelevant.
- If verdict is irrelevant, the key fact must be empty or explicitly state that no useful
  key fact is present.
- Do not fabricate, skip, merge, reorder, or range-compress doc ids.

Stage 2 - Conflict assessment:
Conflict type: <one allowed conflict label>
Reason: <concise conflict_reason, <=50 words>
Evidence pattern: <brief synthesis of how the documents agree, complement, conflict, are
  outdated, contain misinformation, or fail to support an answer>

Stage 2 requirements:
- The conflict type MUST be exactly one of:
  No conflict
  Complementary information
  Conflicting opinions or research outcomes
  Conflict due to outdated information
  Conflict due to misinformation
- Keep the label consistent with the evidence pattern and final answer behavior.

Stage 3 - Answer plan:
<1-2 concise sentences explaining how the cited evidence yields the final answer, or why
  the evidence requires abstention.>

3) A line that is exactly:
</think>

4) ONE BLANK LINE.

5) The FINAL ANSWER lines.

6) A line that is exactly:
[[END-OF-ANSWER]]

FINAL ANSWER REQUIREMENTS
- If you are abstaining: the final answer line must be EXACTLY:
  CANNOT ANSWER, INSUFFICIENT EVIDENCE
  No citations must appear in an abstain answer.

- Otherwise: write 2-4 sentences, or 4-5 for simple unanimous facts.
  - Use bracketed citations [dX]; at least 80% of sentences MUST include at least one
    citation of the form [dX] at the end of that sentence.
  - Cite only existing doc_ids (d1...dN); never cite [dK] where K does not belong to the
    retrieved set.
  - Prefer ordering of cited docs by their credibility (docs you labeled
    source_quality="high" first, then "low"), then by utility.
  - For exact-count questions, prefer a precise count when the retrieved evidence directly
    supports it; otherwise explicitly say the exact count cannot be determined from the
    retrieved evidence.
  - For broad or ambiguous queries, name the scope you are answering within instead of
    implying a universal answer that the snippets do not establish.

No markdown fences, no extra headings, no extra commentary anywhere.
There must be EXACTLY ONE <think>...</think> block (no nesting, no repeats).
The response MUST end with the sentinel line [[END-OF-ANSWER]].

```

Strict user template (user_e2e.txt)

```

You are given a query and retrieved documents.
Use only the retrieved snippets as evidence.

Query:
{query}

Retrieved documents (JSON; `doc_id` is canonical and must be preserved exactly in your
  evidence trace and citations):
{retrieved_docs_json}

```

O.3 RUNTIME

Runtime system prompt (`system_e2e_runtime.txt`)

You are a conflict-aware RAG assistant.

Use only the retrieved documents. Do not use outside knowledge.

Output a public 3-stage evidence trace followed by the final answer:
`<think>`
 Stage 1 - Evidence assessment:
 - dl: supports|partially supports|irrelevant - brief reason grounded in the snippet.

Stage 2 - Conflict assessment:
 Conflict type: one allowed conflict label
 Reason: concise reason
 Evidence pattern: brief synthesis of how the documents agree, complement, conflict, or fail.

Stage 3 - Answer plan:
 Brief plan for the final answer or grounded refusal.
`</think>`

Then write the final answer with end-of-sentence [dX] citations, or exactly:
 CANNOT ANSWER, INSUFFICIENT EVIDENCE

End with `[[END-OF-ANSWER]]`.

Allowed conflict labels: No conflict; Complementary information; Conflicting opinions or research outcomes; Conflict due to outdated information; Conflict due to misinformation.

Abstain only if all documents are irrelevant or the retrieved set still cannot answer the query.
 If any document supports the answer, do not abstain.
 If no single document fully answers the query but multiple partially supporting documents jointly determine the answer, do not abstain; synthesize them carefully.
 Disagreement alone is not a reason to abstain if a grounded conflict-aware answer can still be given.
 Cite only retrieved document ids.

Runtime user template (`user_e2e_runtime.txt`)

Query:
`{query}`

Retrieved documents:
`{retrieved_docs_json}`

O.4 MINIMAL

Minimal system prompt (`system_e2e_minimal.txt`)

You are a RAG assistant.

Answer the query using only the retrieved documents.
 If the documents jointly answer the query, answer it even when each one is only partial.
 Abstain only when the retrieved evidence still leaves a necessary gap.

Minimal user template (`user_e2e_minimal.txt`)

Query:
`{query}`

Retrieved documents:
`{retrieved_docs_json}`

P CATS JUDGE PROMPTS

The complete active judge bundle is reproduced verbatim below: one shared system message, the three task prompts, and the five-type behaviour rubric that the Behaviour Adherence prompt selects

from. Placeholders in braces are filled at evaluation time. Only typographic normalisation was applied.

This is the *entire* bundle. Grounded Refusal has no prompt because it is computed deterministically, and no natural-language-inference prompt appears because the active grounding path does not use one. Earlier NLI-based grounding code exists in the repository but is not the protocol behind any number reported here.

Three properties of these prompts bear on claims made in the main text, and two of them are qualifications rather than confirmations.

Behaviour is explicitly walled off from the other components. The Behaviour Adherence prompt instructs the judge to ignore factual correctness and states outright that answerability correctness, factual entailment, citation validity, unsupported-claim detection and single-truth recall must not be used as hidden criteria. It supplies exactly one rubric line, for the sample’s own conflict type, rather than asking the judge to score all five behaviours. This is the orthogonality requirement of §6 in its operational form. It is corroborated empirically: Behaviour Adherence is the CATS component least correlated with citation behaviour (Appendix I). The prompt also defines a refusal as adherent when the evidence genuinely does not support an answer, which is why correct refusals are excluded from the behaviour denominator rather than scored as failures.

Factual Grounding is deliberately permissive, so it is an upper bound on strict grounding. The prompt enumerates seven distinct routes to support, including paraphrase, entailment, more-specific instances, more-general statements, containing ranges and partial coverage of a claim’s substance. It also instructs the judge to “bias toward inclusion when in doubt”. This is a design choice: the metric is meant to measure whether an answer *uses and cites* retrieved evidence, not whether each claim is strictly entailed. It does mean our grounding numbers should be read as generous rather than conservative, and that a stricter entailment-based variant would score lower.

One wording inaccuracy we do not wish to paper over. The grounding prompt tells the judge that the per-document verdicts it is shown are “ground truth annotations made by human experts”. They are in fact produced by the annotation committee of Appendix B. Human review covered the conflict-type label and the benchmark preselection fields, but *not* the per-document verdicts (Appendix C). The instruction’s operative function is to stop the judge re-litigating document relevance, which it does regardless of provenance, so we do not expect a material effect on the reported grounding scores. But the description is inaccurate, and a corrected wording should be used in any future version.

P.1 SHARED SYSTEM MESSAGE

Committee system message (`committee_json_system_prompt.txt`)

You are an evaluation judge. Respond ONLY with JSON.

P.2 BEHAVIOUR ADHERENCE

Behaviour Adherence prompt (`behavior_adherence_prompt.template.txt`)

You are evaluating ONLY the **behavior** of a model answer, not its factual correctness.

Behavior means:

- How the answer handles multiple sources, uncertainty, disagreement, or lack of conflict.
- Whether it summarizes, reconciles, or contrasts viewpoints as appropriate.
- Whether it is direct vs. hedged, neutral vs. biased, etc.

Given:

- A user query
- A model-generated answer
- A conflict type with ONE expected behavior rubric (shown below)

Your task:

Decide whether the model's answer **follows the expected behavior** for THIS specific conflict type only.

Conflict Type: {conflict_type}

```

Expected Behavior for this conflict type:
{expected_behavior}
{provenance_block}
Instructions:
- Judge the answer ONLY against the rubric shown above. Do not invoke rubrics for other
  conflict types.
- If the answer is a refusal ("CANNOT ANSWER" / "INSUFFICIENT EVIDENCE") AND the retrieved
  evidence genuinely does not support an answer, treat the refusal as adherent.
- If the answer clearly follows the expected behavior for this conflict type, set
  "adherent": true.
- If the answer clearly violates or ignores the expected behavior, set "adherent": false.
- Do not use answerability correctness, factual entailment, citation validity,
  unsupported-claim detection, or single-truth recall as hidden behavior criteria. Those
  are separate metrics.
- Ignore factual correctness; only judge how the answer behaves relative to the rubric.
- The "rationale" should briefly cite the key aspects of the answer's behavior.
- The "confidence" should reflect how certain you are in YOUR adherent decision, between
  0.0 (coin flip) and 1.0 (clearly correct call). Use 0.5 if you are uncertain.

Return ONLY a JSON object with fields:
  "adherent": true or false,
  "rationale": "short explanation",
  "confidence": 0.0 to 1.0

Query:
{query}

Answer:
{answer}

```

Five-type behaviour rubric (behavior_rubric.md)

```

# Behavior Adherence Rubric

This is the exact five-type rubric selected by
`behavior_judge_prompt(...)` in `rag_eval/judge_prompts.py`.

| Conflict type | Rubric supplied to the judge |
| --- | --- |
| 1. No Conflict | Provide a clear and direct answer without introducing alternative
  viewpoints or uncertainty. |
| 2. Complementary Information | Consolidate and reconcile the different partial answers
  provided by the retrieved documents, without framing the response as a debate. |
| 3. Conflicting Opinions or Research Outcomes | Explicitly reflect the debate between the
  retrieved sources and neutrally summarize the different viewpoints. |
| 4. Conflict Due to Outdated Information | Prioritise the up-to-date information, while
  optionally acknowledging the presence of outdated sources. |
| 5. Conflict Due to Misinformation | Disregard inaccurate sources and provide a response
  grounded in reliable and verified information. |

The active prompt selects exactly one rubric for the supplied conflict type. It
does not ask the judge to score all five behaviors at once.

The BA judge is instructed to assess behavior only. It must not use answerability
correctness, factual entailment, citation validity, unsupported-claim detection,
or Single-Truth Recall as hidden BA criteria. A refusal is adherent when the
retrieved evidence genuinely does not support a confident answer, regardless of
conflict type.

```

P.3 FACTUAL GROUNDING

Factual Grounding prompt (factual_grounding_prompt.template.txt)

```

You are a fact-checking judge. Your task is to determine which retrieved documents support
a specific claim from a model's answer.

IMPORTANT GROUND RULES:
- The document verdicts ('supports', 'partially supports') are GROUND TRUTH annotations
  made by human experts. You do NOT need to re-verify whether these docs are relevant to
  the query , they already are. Your job is ONLY to check whether the SPECIFIC CLAIM is
  conveyed by each doc's text.
- Do NOT use outside knowledge. Only what the document text says.
- No concept of 'contradicts' , ignore any conflict between docs. Only look for support.

WHAT COUNTS AS SUPPORT (be generous , partial and paraphrastic support BOTH count):
  A document supports the claim if its text establishes the claim's core assertion. The

```

wording does NOT need to match the claim word-for-word. In particular, the doc supports the claim when ANY of the following holds:

- (a) The doc paraphrases the claim, uses synonyms, or summarizes the same fact.
- (b) The doc states a fact that, taken at face value, entails or implies the claim.
- (c) The doc gives a more specific instance, example, or sub-case of the claim (specific instance -> the general claim is supported).
- (d) The doc gives a more general statement that covers the claim (the claim is a natural sub-case of what the doc says).
- (e) The doc supplies a numeric range, qualitative range, or category that includes the value the claim asserts.
- (f) The doc covers a meaningful portion of the claim's substance , even if secondary modifiers, qualifiers, or peripheral details in the claim aren't in the doc. Missing modifiers do NOT disqualify a doc; missing the core fact does.
- (g) The gold verdict is 'partially supports' AND the doc's key_fact/passage addresses the claim's main assertion , treat this as supporting unless the doc is clearly about a different topic.

WHAT DOES NOT COUNT AS SUPPORT:

- The doc is about a different entity, topic, or time period than the claim.
- The doc only mentions a keyword from the claim without addressing the claim's actual assertion.
- The doc would require external knowledge or several inferential leaps to connect to the claim.

Bias toward INCLUSION when in doubt: if a reasonable reader would say 'yes, this doc backs up the gist of the claim,' mark it supporting. Reserve exclusion for docs that are clearly off-topic or only tangentially related.

QUERY: {query}

{model_answer_block}SPECIFIC CLAIM TO EVALUATE: "{claim_text}"

RETRIEVED DOCUMENTS (pre-annotated supporting docs , decide which contain evidence for the SPECIFIC CLAIM above):
{documents_block}

YOUR TASKS:

1. For each document above, apply the WHAT COUNTS AS SUPPORT rules (a)-(g). List every doc that meets ANY of (a)-(g) in supporting_docs.
2. If NO single document satisfies (a)-(g), check whether the COMBINATION of any two documents together supports the claim (cross-document evidence , one doc provides part A, another provides part B, and together they establish the claim). Only set cross_doc_support=true in this case.

Valid document IDs you may reference: {valid_document_ids}

Return ONLY a valid JSON object with exactly these keys and no comments, no markdown, and no extra text:

```
{
  "supporting_docs": ["d2", "d3"],
  "cross_doc_support": false,
  "cross_doc_combo": []
}
```

Rules for the JSON fields:

- "supporting_docs": doc_ids that individually support the claim; use [] if none.
- "cross_doc_support": true ONLY when no single doc supports but two docs together do.
- "cross_doc_combo": exactly the two doc_ids that combine to support when cross_doc_support is true; otherwise [].

P.4 SINGLE-TRUTH RECALL

Single-Truth Recall prompt (single_truth_recall_prompt.template.txt)

You are checking whether a model answer correctly **asserts** a given factual answer.

The gold answer is the correct answer to a question. Your job is to decide whether the model's answer **commits to** (asserts) this gold answer as the model's own conclusion.

MATCHING RULES (count as adherent=true):

- Paraphrases, abbreviations, and equivalent wording that convey the same fact.
- **Logically equivalent statements** (e.g. "born in 1990" vs "born thirty-five years ago in 2025";

```

"A is the capital of B" vs "B's capital is A").
- Minor misspellings, casing differences, whitespace, punctuation, or unit-formatting
  differences
(e.g. "Chiliwack" ~= "Chilliwack"; "Stephan Curry" ~= "Stephen Curry"; "1,759" ~=
"1759";
"$1.8 billion" ~= "1.8B USD").

NON-MATCHING RULES (count as adherent=false):
- The model commits to a DIFFERENT answer, even if the gold answer also appears somewhere
  in the text
  as a **quotation, citation, or attribution** to a document.
  Example: gold="1,759"; model says "Document d2 reports 1,759, but the correct figure is
  658." ->
  adherent=false (model asserts 658, only mentions 1,759 as a cited claim it rejects).
- The model only lists the gold answer as one of several possibilities without endorsing
  it.
- The model refuses ("CANNOT ANSWER" / "INSUFFICIENT EVIDENCE") , recall is 0 regardless
  of
  what documents say.
- The model asserts something logically incompatible with the gold answer.

Return ONLY a JSON object with fields:
"adherent": true or false (whether the model's answer ASSERTS the gold answer as its own
conclusion),
"rationale": short string explanation (briefly note whether the gold was asserted or
merely mentioned),
"confidence": 0.0 to 1.0 (your certainty in the adherent decision)

Gold factual answer:
{gold_answer}

Model answer:
{model_answer}

```

Q COMPLETE AGREEMENT STATISTICS

The tables in this section report *every* agreement statistic computed in this work, so that no coefficient in the main text is quoted without its denominator, its stratum and its confusion counts. Four conventions apply throughout. Raw agreement and κ answer different questions and are always reported together, because a strongly imbalanced field can show high raw agreement and near-zero κ without either number being wrong. A dash in a κ column marks a field that is single-valued across its stratum, where chance-corrected agreement is undefined. We report exact agreement rather than substituting a coefficient. Nothing is imputed for missing judgments. And committee-internal agreement is agreement among model judges under a fixed prompt and serving protocol, never to be read as human inter-annotator agreement.

Table 29: **Complete resource-review agreement statistics.** Every recorded field in both review workflows. n is the number of review pairs; *agree/disagree* are exact counts. A dash in the κ column marks a field that is single-valued across the stratum, where chance-corrected agreement is undefined and exact agreement is the only meaningful statistic.

Workflow / scope	Field	n	agree	dis.	raw	κ
<i>Training conflict-type review (943 records, 3 reviewers, 2 per item)</i>						
	reviewed_conflict_type	943	787	156	0.835	0.7694
	label_action	943	791	152	0.839	0.1341
	changed_label	943	791	152	0.839	0.1341
	review_confidence	943	607	336	0.644	0.0159
<i>Benchmark preselection (1,454 records, 7 reviewers)</i>						
	human_preselect_decision	1454	1428	26	0.982	0.9228
	preliminary_conflict_type	1454	1378	76	0.948	0.9217
	preselection_confidence	1454	1445	9	0.994	0.9767
	retrieval_quality	1454	1436	18	0.988	0.9550
	evidence_sufficiency	1454	1436	18	0.988	0.9600
	conflict_clarity	1454	1441	13	0.991	0.9660
	query_specificity	1454	1449	5	0.997	0.9875

Workflow / scope	Field	n	agree	dis.	raw	κ
	source_reliability	1454	1446	8	0.994	0.9881
	relevant_doc_count_bin	1454	1426	28	0.981	0.9527
	gold_answer_possible	1454	1440	14	0.990	0.9669
<i>Refusal-quality stratum (128 released refusal items)</i>						
	human_preselect_decision	128	128	0	1.000	–
	preliminary_conflict_type	128	128	0	1.000	1.0000
	preselection_confidence	128	128	0	1.000	–
	retrieval_quality	128	128	0	1.000	–
	evidence_sufficiency	128	128	0	1.000	–
	conflict_clarity	128	128	0	1.000	–
	query_specificity	128	128	0	1.000	–
	source_reliability	128	128	0	1.000	–
	relevant_doc_count_bin	128	128	0	1.000	–
	gold_answer_possible	128	128	0	1.000	–
	refusal_required	128	128	0	1.000	–
	refusal_ground_truth_valid	128	128	0	1.000	–
	refusal_rationale_quality	128	128	0	1.000	–
	refusal_quality_label	128	128	0	1.000	–

Table 29 covers both resource-review workflows. Two features are worth naming. In the training workflow, the retain-or-change and confidence fields show high raw agreement with very low κ . This is a prevalence effect, since most reviewers accepted the committee label and reported high confidence. It is why the five-way taxonomy field is the appropriate headline statistic rather than an average across fields. In the refusal-quality stratum, every field is unanimous across all 128 items, which is why those rows carry exact agreement and no coefficient.

Reviewers are denoted R1–R4 throughout. The mapping is fixed across all tables in this appendix but is otherwise arbitrary and carries no ordering or ranking.

Table 30: **Complete CATS human-evaluation binary agreement.** All strata for all four comparison frames. P_{pos}/P_{neg} are positive- and negative-specific agreement; n_{11} – n_{00} are the confusion cells; α is nominal Krippendorff’s alpha where computed. Conflicting opinions (type 3) is excluded from Single-Truth Recall by construction and so has no STR stratum. Human–human rows are response-level over the 300-response frame; human–committee rows are judgment-level and therefore pool all accepted reviews.

Frame	Metric	Stratum	n	agr	κ	P_{pos}	P_{neg}	n_{11}	n_{10}	n_{01}	n_{00}	α
Human vs human	Behaviour	overall	300	0.853	0.669	0.891	0.776	180	33	11	76	0.667
	Behaviour	R1 / R2	83	0.819	0.587	0.870	0.706	50	14	1	18	0.578
	Behaviour	R1 / R3	84	0.821	0.595	0.870	0.717	50	13	2	19	0.589
	Behaviour	R1 / R4	21	0.952	0.859	0.970	0.889	16	0	1	4	0.862
	Behaviour	R2 / R3	83	0.867	0.719	0.893	0.825	46	5	6	26	0.720
	Behaviour	R2 / R4	19	0.895	0.756	0.923	0.833	12	1	1	5	0.763
	Behaviour	R3 / R4	10	1.000	1.000	1.000	1.000	6	0	0	4	1.000
	Behaviour	type 1 No conflict	63	0.937	0.714	0.964	0.750	53	2	2	6	0.716
	Behaviour	type 2	60	0.783	0.536	0.835	0.683	33	12	1	14	0.522
	Behaviour	Complementary	58	0.862	0.725	0.862	0.862	25	6	2	25	0.727
	Behaviour	type 3	60	0.817	0.505	0.879	0.621	40	8	3	9	0.504
	Behaviour	Conflicting op.	59	0.864	0.725	0.879	0.846	29	5	3	22	0.727
	Behaviour	type 4 Outdated	60	0.817	0.505	0.879	0.621	40	8	3	9	0.504
	Behaviour	type 5	59	0.864	0.725	0.879	0.846	29	5	3	22	0.727
	Behaviour	Misinformation	60	0.817	0.505	0.879	0.621	40	8	3	9	0.504
	STR	overall	215	0.902	0.773	0.929	0.842	138	19	2	56	0.772
	STR	R1 / R2	61	0.902	0.783	0.925	0.857	37	5	1	18	0.784

	Frame	Metric	Stratum	n	agr	κ	P_{pos}	P_{neg}	n_{11}	n_{10}	n_{01}	n_{00}	α
2592													
2593													
2594		STR	R1 / R3	57	0.860	0.672	0.900	0.765	36	8	0	13	0.668
2595		STR	R1 / R4	16	0.938	0.818	0.960	0.857	12	1	0	3	0.823
2596		STR	R2 / R3	61	0.902	0.765	0.930	0.833	40	5	1	15	0.766
2597		STR	R2 / R4	12	1.000	1.000	1.000	1.000	8	0	0	4	1.000
2598		STR	R3 / R4	8	1.000	1.000	1.000	1.000	5	0	0	3	1.000
2599		STR	type 1 No conflict	56	0.911	0.711	0.945	0.762	43	5	0	8	0.710
2600		STR	type 2	43	0.814	0.549	0.871	0.667	27	7	1	8	0.543
2601			Complementary										
2602		STR	type 4 Outdated	57	0.930	0.812	0.953	0.857	41	4	0	12	0.812
2603		STR	type 5	59	0.932	0.865	0.931	0.933	27	3	1	28	0.866
2604			Misinformation										
2605		FG claim	overall	477	0.897	0.780	0.918	0.861	276	25	24	152	0.780
2606		FG claim	R1 / R2	137	0.905	0.786	0.930	0.854	86	0	13	38	0.784
2607		FG claim	R1 / R3	127	0.945	0.888	0.952	0.936	69	4	3	51	0.888
2608		FG claim	R1 / R4	35	0.914	0.821	0.930	0.889	20	0	3	12	0.822
2609		FG claim	R2 / R3	131	0.840	0.663	0.874	0.779	73	21	0	37	0.655
2610		FG claim	R2 / R4	27	1.000	1.000	1.000	1.000	19	0	0	8	1.000
2611		FG claim	R3 / R4	20	0.750	0.519	0.783	0.706	9	0	5	6	0.501
2612		FG claim	type 1 No conflict	77	0.987	0.972	0.990	0.982	49	0	1	27	0.972
2613		FG claim	type 2	109	0.908	0.803	0.928	0.875	64	6	4	35	0.803
2614			Complementary										
2615		FG claim	type 3	115	0.887	0.749	0.914	0.835	69	6	7	33	0.750
2616			Conflicting op.										
2617		FG claim	type 4 Outdated	85	0.800	0.566	0.844	0.721	46	10	7	22	0.568
2618		FG claim	type 5	91	0.912	0.821	0.923	0.897	48	3	5	35	0.821
2619			Misinformation										
2620	Human vs committee	Behaviour	overall	650	0.745	0.407	0.814	0.591	364	68	98	120	–
2621		Behaviour	type 1 No conflict	133	0.865	0.325	0.924	0.400	109	7	11	6	–
2622		Behaviour	type 2	130	0.708	0.279	0.804	0.424	78	4	34	14	–
2623			Complementary										
2624		Behaviour	type 3	128	0.820	0.641	0.816	0.824	51	13	10	54	–
2625			Conflicting op.										
2626		Behaviour	type 4 Outdated	130	0.746	0.338	0.829	0.507	80	19	14	17	–
2627		Behaviour	type 5	129	0.581	0.149	0.630	0.518	46	25	29	29	–
2628			Misinformation										
2629		Behaviour	llama8b	328	0.738	0.355	0.817	0.538	192	39	47	50	–
2630		Behaviour	qwen7b	322	0.752	0.451	0.811	0.636	172	29	51	70	–
2631		STR strict	overall	465	0.908	0.790	0.931	0.858	292	24	19	130	–
2632		STR strict	type 1 No conflict	118	0.924	0.769	0.952	0.816	89	8	1	20	–
2633		STR strict	type 2	94	0.851	0.657	0.891	0.767	57	8	6	23	–
2634			Complementary										
2635		STR strict	type 4 Outdated	124	0.935	0.841	0.955	0.886	85	6	2	31	–
2636		STR strict	type 5	129	0.907	0.814	0.910	0.903	61	2	10	56	–
2637			Misinformation										
2638		STR strict	llama8b	233	0.914	0.801	0.938	0.863	150	11	9	63	–
2639		STR strict	qwen7b	232	0.901	0.779	0.925	0.854	142	13	10	67	–
2640		STR soft	overall	465	0.912	0.798	0.935	0.863	295	21	20	129	–
2641		STR soft	type 1 No conflict	118	0.924	0.762	0.952	0.809	90	7	2	19	–
2642		STR soft	type 2	94	0.872	0.701	0.908	0.793	59	6	6	23	–
2643			Complementary										
2644		STR soft	type 4 Outdated	124	0.935	0.841	0.955	0.886	85	6	2	31	–
2645		STR soft	type 5	129	0.907	0.814	0.910	0.903	61	2	10	56	–
			Misinformation										

Frame	Metric	Stratum	n	agr	κ	P_{pos}	P_{neg}	n_{11}	n_{10}	n_{01}	n_{00}	α
	STR soft	llama8b	233	0.923	0.818	0.944	0.873	153	8	10	62	–
	STR soft	qwen7b	232	0.901	0.779	0.925	0.854	142	13	10	67	–
	FG claim	overall	1029	0.899	0.783	0.920	0.862	599	38	66	326	–
	FG claim	type 1 No conflict	165	0.945	0.880	0.958	0.922	103	4	5	53	–
	FG claim	type 2	233	0.910	0.805	0.929	0.876	138	8	13	74	–
	FG claim	Complementary	250	0.888	0.759	0.911	0.848	144	12	16	78	–
	FG claim	type 3	183	0.825	0.606	0.870	0.733	107	9	23	44	–
	FG claim	Conflicting op.	198	0.929	0.855	0.939	0.917	107	5	9	77	–
	FG claim	type 4 Outdated	183	0.825	0.606	0.870	0.733	107	9	23	44	–
	FG claim	type 5	198	0.929	0.855	0.939	0.917	107	5	9	77	–
	FG claim	Misinformation	518	0.880	0.711	0.916	0.795	336	21	41	120	–
	FG claim	llama8b	511	0.918	0.834	0.926	0.907	263	17	25	206	–
	FG claim	qwen7b	511	0.918	0.834	0.926	0.907	263	17	25	206	–
Human consensus vs committee	Behaviour	overall	256	0.789	0.483	0.852	0.630	156	24	30	46	–
	Behaviour	type 1 No conflict	59	0.881	0.299	0.935	0.364	50	3	4	2	–
	Behaviour	type 2	47	0.787	0.391	0.865	0.500	32	1	9	5	–
	Behaviour	Complementary	50	0.880	0.760	0.885	0.875	23	2	4	21	–
	Behaviour	type 3	49	0.796	0.419	0.868	0.545	33	7	3	6	–
	Behaviour	Conflicting op.	51	0.588	0.165	0.632	0.533	18	11	10	12	–
	Behaviour	type 4 Outdated	49	0.796	0.419	0.868	0.545	33	7	3	6	–
	Behaviour	type 5	51	0.588	0.165	0.632	0.533	18	11	10	12	–
	Behaviour	Misinformation	194	0.948	0.876	0.964	0.912	132	6	4	52	–
	STR strict	overall	51	0.961	0.865	0.976	0.889	41	2	0	8	–
	STR strict	type 1 No conflict	35	0.914	0.767	0.943	0.824	25	2	1	7	–
	STR strict	type 2	53	0.962	0.898	0.975	0.923	39	2	0	12	–
	STR strict	Complementary	55	0.945	0.891	0.947	0.943	27	0	3	25	–
	STR strict	type 4 Outdated	53	0.962	0.898	0.975	0.923	39	2	0	12	–
	STR strict	type 5	55	0.945	0.891	0.947	0.943	27	0	3	25	–
	STR strict	Misinformation	194	0.954	0.888	0.967	0.920	133	5	4	52	–
	STR soft	overall	51	0.961	0.865	0.976	0.889	41	2	0	8	–
	STR soft	type 1 No conflict	35	0.943	0.838	0.963	0.875	26	1	1	7	–
	STR soft	type 2	53	0.962	0.898	0.975	0.923	39	2	0	12	–
	STR soft	Complementary	55	0.945	0.891	0.947	0.943	27	0	3	25	–
	STR soft	type 4 Outdated	53	0.962	0.898	0.975	0.923	39	2	0	12	–
	STR soft	type 5	55	0.945	0.891	0.947	0.943	27	0	3	25	–
	STR soft	Misinformation	428	0.949	0.888	0.960	0.928	264	12	10	142	–
	FG claim	overall	76	0.947	0.885	0.959	0.926	47	2	2	25	–
	FG claim	type 1 No conflict	99	0.949	0.890	0.961	0.930	61	3	2	33	–
	FG claim	type 2	102	0.961	0.912	0.971	0.941	66	3	1	32	–
	FG claim	Complementary	68	0.897	0.762	0.925	0.837	43	3	4	18	–
	FG claim	type 3	83	0.976	0.951	0.979	0.971	47	1	1	34	–
	FG claim	Conflicting op.	68	0.897	0.762	0.925	0.837	43	3	4	18	–
	FG claim	type 4 Outdated	83	0.976	0.951	0.979	0.971	47	1	1	34	–
	FG claim	type 5	83	0.976	0.951	0.979	0.971	47	1	1	34	–
	FG claim	Misinformation	428	0.949	0.888	0.960	0.928	264	12	10	142	–
Human vs individual judge	Behaviour	local/deepseek-r1-distill-32b	650	0.735	0.339	0.820	0.503	391	41	131	87	–
	Behaviour	local/mistral-small-4	650	0.643	0.229	0.720	0.506	299	133	99	119	–
	Behaviour	local/qwen3.5-397b-a17b	649	0.743	0.405	0.812	0.592	361	70	97	121	–

Table 30 gives all four binary comparison frames at every stratum, including the six reviewer pairs, the five conflict types, the per-model strata and the per-judge comparison, with positive- and negative-specific agreement and the full confusion cells. The reviewer-pair rows show that agreement is not uniform across annotators either: human behaviour agreement ranges from 0.819 to 1.000 across pairs. The smallest pairs involve the reviewer who completed half an assigned queue, so their high coefficients rest on very few items and should not be read as evidence of superior reliability.

Table 31: **Complete continuous grounding-ratio agreement.** Per-sample Factual-Grounding ratios compared as continuous values. *exact* uses the configured tolerance rather than floating-point identity. Correlation measures association, not calibration, so it must be read with MAE/RMSE and the two means. Human-human rows are response-level over the 300-response frame; human-committee rows are judgment-level.

Frame	Stratum	<i>n</i>	<i>exact</i>	MAE	RMSE	Pearson	Spearman	mean A	mean B
Human vs human	overall	300	0.860	0.096	0.278	0.807	0.796	0.647	0.635
	R1 / R2	83	0.855	0.074	0.224	0.887	0.871	0.601	0.676
	R1 / R3	84	0.929	0.062	0.236	0.865	0.869	0.617	0.627
	R1 / R4	21	0.905	0.048	0.163	0.937	0.923	0.647	0.694
	R2 / R3	83	0.783	0.169	0.383	0.693	0.686	0.738	0.569
	R2 / R4	19	1.000	0.000	0.000	1.000	1.000	0.697	0.697
	R3 / R4	10	0.600	0.233	0.408	0.641	0.638	0.433	0.667
	type 1 No conflict	63	0.984	0.016	0.126	0.965	0.965	0.640	0.656
	type 2	60	0.850	0.092	0.263	0.826	0.798	0.672	0.625
	Complementary								
	type 3 Conflicting op.	58	0.793	0.135	0.329	0.696	0.680	0.645	0.654
	type 4 Outdated	60	0.783	0.161	0.368	0.661	0.642	0.694	0.633
	type 5	59	0.881	0.079	0.251	0.855	0.862	0.583	0.606
	Misinformation								
Human vs committee	overall	650	0.865	0.095	0.281	0.805	0.798	0.627	0.655
	type 1 No conflict	133	0.932	0.050	0.207	0.901	0.885	0.657	0.662
	type 2	130	0.854	0.109	0.305	0.751	0.754	0.633	0.660
	Complementary								
	type 3 Conflicting op.	128	0.797	0.118	0.294	0.765	0.752	0.604	0.654
	type 4 Outdated	130	0.815	0.146	0.356	0.687	0.677	0.651	0.685
	type 5	129	0.922	0.054	0.213	0.897	0.894	0.588	0.614
	Misinformation								
	llama8b	328	0.845	0.109	0.299	0.766	0.755	0.675	0.713
	qwen7b	322	0.885	0.082	0.261	0.836	0.832	0.578	0.596
Consensus vs committee	overall	258	0.938	0.045	0.195	0.909	0.906	0.648	0.636
	type 1 No conflict	62	0.935	0.046	0.195	0.914	0.897	0.651	0.648
	type 2	51	0.922	0.069	0.252	0.836	0.844	0.650	0.641
	Complementary								
	type 3 Conflicting op.	46	0.935	0.025	0.101	0.972	0.974	0.654	0.643
	type 4 Outdated	47	0.894	0.089	0.280	0.816	0.805	0.681	0.642
	type 5	52	1.000	0.000	0.000	1.000	1.000	0.606	0.606
	Misinformation								

Table 31 reports the continuous grounding-ratio comparisons. These are complementary to the claim-level statistics rather than redundant. Claim-level agreement asks whether reviewers make the same individual support decisions, while the ratio compares the proportion of an answer they judge grounded. A reviewer pair can agree on most claims yet differ on the ratio when their extracted claim sets differ in size.

Table 32: **Complete committee-internal agreement.** Agreement among the three judge models on the 300-response agreement frame. This is agreement among model judges under a fixed prompt and serving protocol; it is not human inter-annotator agreement and must not be reported as such.

Subset	Metric	Judge pair	n	agr	κ	P_{pos}	P_{neg}
300-response frame	Behaviour	<i>alpha / mean pairwise</i>	300	0.4568 / 0.4699			
		DeepSeek-32B vs Mistral-4	300	0.730	0.368	0.810	0.532
		DeepSeek-32B vs Qwen-397B	300	0.853	0.611	0.903	0.703
	STR strict	Mistral-4 vs Qwen-397B	300	0.743	0.431	0.806	0.621
		<i>alpha / mean pairwise</i>	215	0.6780 / 0.6874			
		DeepSeek-32B vs Mistral-4	215	0.809	0.613	0.839	0.766
	FG claim	DeepSeek-32B vs Qwen-397B	215	0.944	0.874	0.958	0.915
		Mistral-4 vs Qwen-397B	215	0.791	0.575	0.825	0.740
		<i>alpha / mean pairwise</i>	477	0.8706 / 0.8705			
		DeepSeek-32B vs Mistral-4	477	0.950	0.886	0.963	0.924
		DeepSeek-32B vs Qwen-397B	477	0.941	0.869	0.956	0.913
		Mistral-4 vs Qwen-397B	477	0.937	0.857	0.953	0.903

Table 32 reports committee-internal agreement on the 300-response frame. The reliability ordering is clear: grounding is high ($\alpha \approx 0.87$), recall intermediate (≈ 0.67), behaviour lowest (≈ 0.46). The behaviour disagreement is concentrated between specific judge pairs rather than spread evenly, which is consistent with the per-judge human comparison in Table 21.

R COMPLETE RESULTS MATRIX

Refusal recall moves from 0.000–0.633 to 0.898–1.000, and several fine-tuned models correctly abstain on all 128 refusal-required items. Baseline Llama-3.1-8B under the minimal prompt shows why decision accuracy and refusal F1 must be read together: its decision accuracy of 0.825 looks respectable, but the model correctly refuses one of 128 refusal-required items, visible only in its refusal F1 of 0.015. The per-family Behaviour Adherence gains are +0.096 (Mistral-7B), +0.068 (Llama-3.1-8B), +0.047 (Qwen2.5-32B) and +0.000 (Qwen2.5-7B). The Qwen baselines already score 0.789 and 0.811 under the strict prompt against 0.660 for Mistral-7B, leaving less room for supervision to add.

Table 33 gives all twelve model×prompt cells, base and fine-tuned.

S LIMITATIONS, EXPANDED

The main-text Limitations section (§10) keeps the three caveats that bound what our central claims mean. This appendix gives six further, narrower limitations trimmed from the main text for space.

Shared model families between annotation and evaluation. The annotation committee and the evaluation committee draw on the same three open-weight model families, with different weightings. The constructs differ (document adjudication versus answer judging), and Grounded Refusal is deterministic and outside the committee entirely. But the evaluator is partly scoring labels produced by related models. The human study in §9 is our external check on this and it covers two of the four model families.

Aggregate properties. The type-balanced aggregate assigns 40% of its weight to refusal-required examples, which score decision correctness only, against 17.4% of the data. One type-3 refusal example carries roughly twelve times the influence of one type-1 answerable example. And the misinformation class, 5% of the data, determines 20% of the score. A model with zero usable answer content but perfect refusal calibration would score 0.400. This is why we report components first and aggregates second (Appendix G.8).

Model	Prompt	Run	Dec-Acc	Ref-F1	BA	FG	STR	AQ	CATS-P	CATS-B
Qwen2.5-7B	strict	base	0.707	0.429	0.811	0.617	0.561	0.524	0.450	0.459
Qwen2.5-7B	strict	SFT	0.967	0.911	0.780	0.838	0.687	0.677	0.630	0.684
Qwen2.5-7B	runtime	base	0.784	0.361	0.768	0.651	0.648	0.541	0.453	0.375
Qwen2.5-7B	runtime	SFT	0.966	0.906	0.774	0.829	0.693	0.690	0.629	0.679
Qwen2.5-7B	minimal	base	0.827	0.031	0.737	0.031	0.757	0.033	0.022	0.017
Qwen2.5-7B	minimal	SFT	0.958	0.881	0.768	0.837	0.686	0.680	0.617	0.659
Qwen2.5-32B	strict	base	0.855	0.367	0.789	0.851	0.733	0.724	0.574	0.432
Qwen2.5-32B	strict	SFT	0.963	0.905	0.821	0.812	0.749	0.686	0.661	0.707
Qwen2.5-32B	runtime	base	0.855	0.389	0.799	0.864	0.780	0.760	0.600	0.455
Qwen2.5-32B	runtime	SFT	0.962	0.901	0.804	0.814	0.758	0.701	0.658	0.716
Qwen2.5-32B	minimal	base	0.826	0.015	0.747	0.108	0.773	0.105	0.086	0.057
Qwen2.5-32B	minimal	SFT	0.966	0.911	0.809	0.822	0.738	0.687	0.652	0.716
Llama-3.1-8B	strict	base	0.762	0.255	0.721	0.705	0.634	0.577	0.426	0.363
Llama-3.1-8B	strict	SFT	0.973	0.926	0.805	0.852	0.727	0.706	0.680	0.744
Llama-3.1-8B	runtime	base	0.826	0.030	0.751	0.897	0.720	0.746	0.510	0.335
Llama-3.1-8B	runtime	SFT	0.974	0.930	0.806	0.849	0.705	0.690	0.673	0.748
Llama-3.1-8B	minimal	base	0.825	0.015	0.743	0.074	0.710	0.055	0.029	0.018
Llama-3.1-8B	minimal	SFT	0.976	0.934	0.795	0.857	0.714	0.694	0.672	0.732
Mistral-7B	strict	base	0.845	0.352	0.660	0.624	0.563	0.489	0.378	0.303
Mistral-7B	strict	SFT	0.927	0.826	0.793	0.796	0.644	0.632	0.620	0.677
Mistral-7B	runtime	base	0.826	0.200	0.676	0.739	0.590	0.566	0.398	0.299
Mistral-7B	runtime	SFT	0.929	0.831	0.760	0.762	0.613	0.593	0.587	0.657
Mistral-7B	minimal	base	0.823	0.000	0.727	0.079	0.700	0.073	0.048	0.025
Mistral-7B	minimal	SFT	0.933	0.839	0.786	0.792	0.627	0.620	0.608	0.663

Table 33: **End-to-end results, all twelve model×prompt cells** (24 rows). Dec-Acc is decision accuracy; GR-P/R/F1 are answer-positive; Ref-F1 is refusal-positive. $n = 736$ throughout. Oracle-condition rows are summarised in Table 27.

Recipe asymmetry and scale. Qwen models use recipe K and Llama/Mistral recipe L. No completed Qwen matrix exists for recipe L. Recipes J/K/L change several ingredients together and are principled recipe revisions rather than single-factor ablations. All models are in the 7B–32B range and Mistral-7B remains over-abstaining under every recipe we tried, a model-family limitation we report in full.

Annotation quality. Supporting quotes are snippet-*anchored* rather than fully verbatim: 81–83% are exact contiguous substrings and ~96% are traceable to the snippet, with ~4% showing no detectable overlap. Source-quality labels come from an explicit URL whitelist rule (Appendix B), not from an independent credibility assessment, and are systematically pushed toward *low* on the refusal track, whose upstream source uses placeholder URLs. Three benchmark records carry a numeric conflict identifier inconsistent with their authoritative textual label. We verified this did not affect any reported result.

Human review coverage. The benchmark second review pass is sequential rather than blind, so its agreement figures measure verification consistency. Of the 943 consolidated training rows, 285 are consensus-completed instead of genuine two-reviewer pairs and 105 of the 608 answerable benchmark records appear only as consensus-completed entries. Several funnel stages (1,878 assigned candidates against 1,221 reviewed; 200 curated refusals against 128 released) have no retained per-record selector. We report the counts as observed rather than inferring a rule.

Comparator scope. The prompting comparators cover three of the four model families and one prompt configuration each. They are not exhaustive prompt-engineering baselines. One comparator metric shifted substantially between our earlier evaluation and this one, in a way most consistent with a corrected answer-extraction step. We therefore base the comparator conclusion on abstention behaviour and citation hygiene, which are stable across both, rather than on grounding alone. The answer-only SFT comparator (§8.6, Appendix E) covers two of the four model families (Llama-3.1-

2862 8B, Qwen2.5-7B) and its own two-prompt ladder, narrower than the four-model, three-prompt matrix
2863 reported elsewhere.
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